

NATIONAL UNIVERSITY OF PAKISTAN



MODEL PAPERS

BS PROGRAM (SEMESTER - II)





Model Papers (Semester-II)

Spring-2025



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GENERAL COURSES



NATIONAL UNIVERSITY OF PAKISTAN
BS Program (Model Paper)
End-Semester Examination

Reg.No: _____

Course Code: GEN-104

Course Title: Expository Writing

Sign of Invigilator: _____

Time Allowed: 20 minutes

Total Marks: 10

General Instructions

Deleting/Overwriting is not allowed. Do not use lead pencil.

All parts of objective section are to be answered on this paper and be handed in to the invigilator.

SECTION – A**Q1. Choose the most appropriate option.****(10x1=10)**

- i. Expository writing _____.
a. informs b. tries to persuade c. persuades d. is subjective
- ii. By reading critically, we can comprehend _____ meaning.
a. literal b. textual c. surface d. hidden
- iii. Questioning is a _____ strategy.
a. writing b. reading c. listening d. speaking
- iv. Quaid was born on 25th December. It is a/an _____.
a. inference b. opinion c. fact d. none of the above
- v. _____ is a strategy of generating ideas.
a. brainstorming b. mind maps c. charts d. a, b, c
- vi. She studied hard; therefore she was fairly confident about her chances. The underlined is
a. preposition b. adjective c. noun d. cohesive device
- vii. Ethics calls for avoiding _____ in expository writing.
a. plagiarism b. verbosity c. brevity d. visual material
- viii. The tone of expository writing is _____.
a. academic b. objective c. neutral d. a, b, c
- ix. In writing, the choice of words based on audience, context and situation is _____.
a. tone b. voice c. diction d. style
- x. A/an _____ essay explores multiple perspectives on a topic and presents a balanced view.
a. descriptive b. argumentative c. narrative d. discursive

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: GEN-104**Course Title:** Expository Writing**Sign of Invigilator:** _____**Time Allowed:** 2 Hours 40 Minutes**Total Marks:** 40**General Instructions**

- Do not write anything on the question paper other than Registration No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

SECTION – B**Q.2 Do as directed:****(5x2=10)**

- How is Expository Writing different from Persuasive Writing?
- What role do Cohesive Devices play in achieving coherence in a paragraph?
- How can Visual Communication be made effective?
- What are the challenges in AI-assisted writing?
- How can Expository Writing be adapted for Digital Platforms?

SECTION – C**Write a detailed answer to the following:****(10x3=30)****Q.3. a.** The first stage in Writing Process is to generate ideas. How is it done? **(05)**

b. Expository writing often involves presenting information from various sources. What ethical considerations should be kept in mind when using sources and how can academic integrity be ensured? **(05)**

Q.4. *You are the Principal of a College and you are required to dispatch Exam Registration Forms (Spring '25) to the Controller Exam NUP. Write a Covering Letter for the dispatch. **(10)**

Q.5. What are the common and distinctive characteristics of argumentative and discursive paragraphs? **(10)**

*Note: Resume Writing, Email Writing and Paragraph Writing may also be given.

**Answer Key**

QUESTION NUMBER	ANSWER
1	A
2	D
3	D
4	C
5	D
6	D
7	A
8	D
9	C
10	D

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: GEN-105**Course Title: Quantitative Reasoning-II****Sign of Invigilator: _____****Time Allowed: 20 minutes****Total Marks: 10****General Instructions**

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A**Q1. Choose the most appropriate option.****(10*1=10)**

- i. There are 1029 students in a school. 504 of them are girls. What will be the ratio of boys to the girls?
- a) 25:24 b) 20:19 c) 24:25 d) 18:19
- ii. A motorcycle covers 90km. in 2litres of petrol. In how many liters of petrol will it cover 225km.?
- a) 9 litres b) 8 litres c) 6 litres d) 5 litres
- iii. If $x^2 + 7x + 12 = 0$. Then solution set will be
- a) $\{-3, -4\}$ b) $\{3, 4\}$ c) $\{2, 3\}$ d) $\{1, 2\}$
- iv. Perimeter of circle is
- a) πr b) $2\pi r$ c) 4π d) $\frac{4}{3}\pi r^3$
- v. The area of triangle is $90m^2$. If base is 15meter then altitude will be?
- a) 14meter b) 18meter c) 12meter d) 16meter
- vi. Solution set of $3x + 5 < x - 7$
- a) $]-\infty, -6[$ b) $]6, \infty[$ c) $]-\infty, -7[$ d) $]7, \infty[$
- vii. If $a_{n-2} = 3n - 11$ then a_2 will be
- a) 2 b) 4 c) 1 d) 6
- viii. The general term of a G.P is
- a) $a_1 + (n - 1)d$ b) $a_1 + nd$ c) $a_1 r^n$ d) $a_1 r^{n-1}$
- ix. If the cost price of an article is \$250 and it is sold for \$300, what is the profit percentage?
- a) 15% b) 20% c) 25% d) 30%
- x. If the 9th term of H.P is $1/5$. Then the 9th term of A.P is
- a) 8 b) $1/5$ c) 5 d) 10

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)****End-Semester Examination**

Reg.No: _____

Course Code: GEN-105**Time Allowed:** 02 Hrs and 40 minutes**Course Title:** Quantitative Reasoning-II**Total Marks:** 40**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

SECTION – B**Q.2 Give short answers to the following questions:****(5*2=10)**

- Define types of proportion.
- Solve for x, $\frac{c}{cx-1} + \frac{b}{bx-1} = c + b$; $x \neq \frac{1}{c}, \frac{1}{b}$
- The diameter of a semi-circular shape is 14cm. What will be the area of this shape?
- Find the solution set of $|5x + 6| \geq 5$.
- If $\frac{1}{a}, \frac{1}{b}$ and $\frac{1}{c}$ are in Arithmetic Progression, show that $= \frac{2ac}{a+c}$.

SECTION – C**Write detailed answers to the following questions.****(10*3=30)**

Q.3 a) An airplane carries 400 passengers, 52% of the passengers were Pakistani, 17% were Chinese, 12% were from Iran and the rest were British. (5)

i) How many of each nationality were on the plane?

ii) What percentage were British?

b) Show that roots of a quadratic equation $x^2 + (mx + c)^2 = a^2$ will be equal if $c^2 = a^2(1 + m^2)$.

(5)

Q.4 a) If α and β are roots of $5x^2 - x - 2 = 0$, form the equation whose roots are $\frac{3}{\alpha} + \frac{3}{\beta}$. (5)

b) The length of a rectangular garden is one meter more than its width if the perimeter of garden is 12 meters find the length and width of the garden.

Q. 5 a) Solve $|z + 3| = 6 - |z + 3|$. (5)

b) Find the n th term of the Geometric sequence if $\frac{a_5}{a_3} = \frac{4}{9}$ and $a_2 = \frac{4}{9}$. (5)

**Answer Key**

QUESTION NUMBER	ANSWER
1	A
2	D
3	A
4	B
5	C
6	A
7	C
8	D
9	B
10	C

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: GEN-106**Time Allowed: 20 minutes****Course Title: Islamic Studies****Total Marks: 10****Sign of Invigilator: _____****General Instructions**

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A**Q1. Choose the most appropriate option.****(10x1=10)**

- I. About whom did the Prophet ﷺ say that this son of mine is a leader?
 - (a) Hazrat Hassan (b) Hazrat Hussain (c) Hazrat Ali (d) Hazrat Zaid
- II. The Pledge of Ridwan was taken to avenge the murder of which companion?
 - (a) Hazrat Ali (b) Hazrat Abu Bakr (c) Hazrat Umar (d) Hazrat Usman
- III. At which age a child is ordered to offer prayer?
 - (a) Six (b) Seven (c) Eight (d) Nine
- IV. The first Islamic state was founded in?
 - (a) in Madinah (b) in Makkah (c) in Taif (d) in Kufa
- V. What is the meaning of النَّجُومُ?
 - (a) Sun (b) Moon (c) Stars (d) Clouds
- VI. Who is the most honorable in the sight of Allah?
 - (a) Rich (b) Honest businessman (c) Pious (d) Good morals
- VII. When did the Pact of Al-Hudaybiyah?
 - (a) 7 h (b) 6 h (c) 5 h (d) 4 h
- VIII. How many daughters did the Prophet ﷺ have?
 - (a) 4 (b) 3 (c) 5 (d) 2
- IX. In Surah Al-Ahzab, which people has Allah cursed in this world and the hereafter?
 - (a) The hypocrites (b) The disbelievers (c) The polytheists (d) Those who hurt Allah and His Messenger
- X. Which Companion was given the title of "الامت امين".
 - (a) Hazrat Umar (b) Hazrat Abu Ubaydah Bin-al-Jarah (c) Hazrat Ali (d) Hazrat Abu Bakr

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: GEN-106**Time Allowed: 02 Hrs and 40 minutes****Course Title: Islamic Studies****Total Marks: 40****Sign of Invigilator: _____****General Instructions**

- Do not write anything on the question paper other than Registration No.
- Attempt all questions. All the Q&A must be numbered as mentioned in the question paper.

SECTION – B**Q.2: Write short answers to the following.****(5*2 = 10)**

- Write the two attributes of the pious mentioned in the beginning of Surah Al-Baqarah.
- Write any two social manners mentioned in Surah Al-Hujarat.
- Write any two terms of Treaty of Madinah.
- What are the five things that will be questioned first on the Day of Judgment?
- Write two main points of the last sermon of the Holy Prophet.

SECTION – C**Write detailed answers to the following.****(10*3=30)****Q.3: I. Translate and explain the following verse.****(5)**

كَثِيرًا اللَّهُ وَذَكَرَ الْآخِرَ وَالْأَوَّلَ يَوْمَ اللَّهُ يَرَّجُوا كَانَ لِمَنْ حَسَنَةً أَسْوَدَةٌ اللَّهُ رَسُولٌ فِي لَكُمْ كَانَ لَقَدْ

II. Translate and explain the following Hadith.**(5)**يُبْغِضُ تَعَالَى اللَّهُ، فَإِنَّ حَسَنَ خَلْقٍ مِنْ الْقِيَامَةِ يَوْمَ الْمُؤْمِنِ مِيزَانٍ فِي أَثْقَلِ شَيْءٍ مِنْ مَا وَسَلَّمَ عَلَى اللَّهِ صَلَّى اللَّهُ رَسُولٌ قَالَ
الْبُذِي الْفَاحِشَ**Q.4 Explain the need and importance of studying the Seerah of the Holy Prophet (10)****Q. 5 Write the characteristics of Islamic civilization and culture.****(10)**



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN**BS Program (Model Paper)****End-Semester Examination****Course Code: GEN-106****Time Allowed: 20 minutes****Course Title: Islamic Studies****Total Marks: 10****Sign of Invigilator: _____**

ہدایات: حذف کر کے / کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ براہ کرم لیڈ پینسل کا استعمال نہ کریں۔

حصہ اول (حصہ معروضی) کے تمام اجزاء کا جواب اسی پرچے پر دیں۔

حصہ اول

سوال نمبر 1- مناسب جواب کا انتخاب کریں۔

- i. نبی ﷺ نے کس کے بارے میں فرمایا کہ میرا یہ بیٹا سردار ہے؟
 (ا) حضرت حسنؓ (ب) حضرت حسینؓ (ج) حضرت علیؓ (د) حضرت زیدؓ
- ii. بیعت رضوان کس صحابی کے خون کا بدلہ لینے کے لیے کی گئی؟
 (ا) حضرت علیؓ (ب) حضرت ابوبکرؓ (ج) حضرت عمرؓ (د) حضرت عثمانؓ
- iii. بچہ کتنے برس کا ہو تو اسے نماز کا حکم دو؟
 (ا) چھ (ب) سات (ج) آٹھ (د) نو
- iv. پہلی اسلامی ریاست کی بنیاد رکھی گئی؟
 (ا) مدینہ میں (ب) مکہ میں (ج) طائف میں (د) کوفہ میں
- v. النجوم کا کیا معنی ہے؟
 (ا) سورج (ب) چاند (ج) ستارے (د) بادل
- vi. اللہ کے نزدیک عزت والا کون ہے؟
 امیر (۱) (ب) ایماندار تاجر (ج) متقی (د) اچھے اخلاق والا
- vii. صلح حدیبیہ کب ہوئی؟
 (ا) 7ھ (ب) 6ھ (ج) 5ھ (د) 4ھ
- viii. نبی ﷺ کی کتنی بیٹیاں ہیں؟
 (ا) 4 (ب) 3 (ج) 5 (د) 2
- ix. سورہ احزاب میں کن لوگوں پر اللہ نے دنیا و آخرت میں لعنت کی ہے؟
 (ا) منافقین پر (ب) کافروں پر (ج) مشرکین پر (د) اللہ اور رسول ﷺ کو ایذا دینے والوں پر
- x. امین الامت "کس صحابی کا لقب ہے؟"
 (ا) حضرت عمرؓ (ب) حضرت ابو عبیدہ بن جراحؓ (ج) حضرت علیؓ (د) حضرت ابو بکرؓ

**NATIONAL UNIVERSITY OF PAKISTAN**

BS Program (Model Paper)

End-Semester Examination

Reg.No: _____

Course Code: GEN-106

Time Allowed: 02 Hrs and 40 minutes

Course Title: Islamic Studies

Total Marks: 40

Sign of Invigilator: _____

ہدایات: سوالیہ پرچے پر رول نمبر کے علاوہ کچھ نہ لکھیں۔ تمام سوالات اور جوابات کو سوالیہ پرچے کے مطابق نمبر دیں۔

حصہ دوم

(5*2 = 10)

سوال نمبر 2 - درج ذیل سوالات کے مختصر جوابات تحریر کریں

- i-سورة البقرہ کے آغاز میں بیان کردہ متقین کی دو صفات تحریر کریں -
- ii-سورة الحجرات میں بیان کیے گئے کوئی سے دو معاشرتی آداب تحریر کریں۔ -
- iii-میثاق مدینہ کی کوئی سی دو شرائط لکھیں -
- iv-روز قیامت سب سے پہلے کن پانچ چیزوں کے بارے میں سوال کیا جائے گا۔ -
- v-خطبہ حجة الوداع کے دو بنیادی نکات بیان کیجئے۔ -

حصہ سوم

(10*3=30)

سوال نمبر 2 - درج ذیل سوالات کے تفصیلی جوابات

تحریر کریں

آیت کا ترجمہ و تشریح لکھیں۔

(5)

لَقَدْ كَانَ لَكُمْ فِي رَسُولِ اللَّهِ أُسْوَةٌ حَسَنَةٌ لِّمَن كَانَ يَرْجُوا اللَّهَ وَالْيَوْمَ
الْآخِرَ وَذَكَرَ اللَّهَ كَثِيرًا

(5)

حدیث کا ترجمہ و تشریح لکھیں۔

قال رسول الله صلى الله عليه وسلم "ما من شيء أثقل في ميزان المؤمن يوم القيامة من خلق حسن، فإن الله تعالى يفيض الفاحش البذي،

سوال نمبر 4- مطالعہ سیرت النبی ﷺ کی ضرورت و

(10)

اہمیت بیان کیجئے۔

سوال نمبر 5 - اسلامی تہذیب و ثقافت کے خصائص لکھیں۔

(10)

**Answer Key**

QUESTION NUMBER	ANSWER
1	A
2	D
3	B
4	A
5	C
6	C
7	B
8	A
9	D
10	B

QUESTION NUMBERS	ANSWERS
1	ا
2	د
3	ب
4	ا
5	ج
6	ج
7	ب
8	ا
9	د
10	ب

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: GEN-107**Course Title: Environmental Science****Sign of Invigilator: _____****Time Allowed: 20 minutes****Total Marks: 10****General Instructions**

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION-A

Q1. Choose the most appropriate option.

(10x1=10)

- Which environment issue is associated with the depletion of the ozone layer?
 - Carbon mono oxide emissions
 - Sulphur dioxide emissions
 - Nitrogen gas emission
 - Emission of chloroform carbons (CFCs)
- The National Environmental Quantity standards (NEQs) in Pakistan aim to
 - Promote industrial growth
 - Regulate environmental pollutants
 - Encourage deforestation
 - Increase carbon emission
- Which of the following is an example of a non-renewable source?
 - Wind energy
 - Geo-thermal energy
 - Natural Gas
 - Bio mass
- Electrostatic precipitators are used to control:
 - Water pollution
 - Soil erosion
 - Noise pollution
 - Air pollution
- Agreement which focuses on the transboundary movement of hazardous waste is
 - Basal Convention
 - Montreal protocol
 - Paris Agreement
 - Ramsar Convention
- The term sustainable development refers to
 - Development without considering future generations
 - Development that meets present needs without compromising future generation's ability to meet their own needs.
 - Rapid industrialization
 - Economic growth at any environmental cost.



- vii.** The primary source of phosphorous in an ecosystem is
a. Atmosphere b. Ocean water c. Fossil fuels d. Rocks and minerals
- viii.** The variety of life in biosphere refers to
a. Biodiversity b. Ecosystem c. Biogeochemical cycle d. Food web
- ix.** Which of the following is an example of a terrestrial ecosystem?
a. Coral reef b. Lake c. Desert d. Ocean
- x.** The layer which lies beneath the lithosphere is
a. Asthenosphere b. Outer Core c. Inner Core d. Mesosphere



NATIONAL UNIVERSITY OF PAKISTAN

BS Program (Model Paper)
End-Semester Examination

Reg.No: _____

Course Code: GEN-107

Time Allowed: 02 Hrs and 40 minutes

Course Title: Environmental Science

Total Marks: 40

Sign of Invigilator: _____

General Instructions

- Do not write anything on the question paper other than Registration No.
- Attempt all questions. All the Q&A must be numbered as mentioned in the question paper.

SECTION – B

Q.2 Write short answers to the following questions:

(5x2=10)

- What are the environmental impacts of plastic pollution on terrestrial ecosystems?
- Discuss the role of the Montreal Protocol in protecting the ozone layer.
- Discuss the significance of agro ecology in promoting sustainable agriculture.
- Describe the primary functions of Earth's atmosphere and explain its role in environmental systems?
- Sketch Nitrogen cycle.

SECTION – C

Write detailed answers of the following questions.

(3x10=30)

Q.3. Analyze the environmental and socio-economic impacts of transboundary water pollution, using the example of the Kabul River Basin, suggest strategies for effective management. **(10)**

Q.4. Explain the concept of plastisphere and its significance in understanding microbial interactions with micro plastics. What are the potential risks of micro plastic pollution's human health and the environment?

(10)

Q.5 Discuss the causes and consequences of climate change. How can sustainable development goals address these challenges? **(10)**

**Answer Key**

QUESTION NUMBER	ANSWER
1	D
2	B
3	C
4	D
5	A
6	B
7	D
8	A
9	C
10	A



NATIONAL UNIVERSITY OF PAKISTAN

BS Program (Model Paper)
End-Semester Examination

Reg.No: _____

Course Code: GEN-109

Course Title: Sociology

Sign of Invigilator: _____

Time Allowed: 20 minutes

Total Marks: 10

General Instructions

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A

Q1. Choose the most appropriate option.

(10*1=10)

1. Why is language considered an important element of culture?

- A) Because it is only a tool for writing
- B) Because it is used only on formal occasions
- C) Because it helps in the transmission of ideas, knowledge, and traditions
- D) Because it is important only for specific groups of people answer is

2. What is the best definition of culture?

- A) The material objects used in a society
- B) The knowledge, beliefs, morals, customs, and habits acquired and shared by members of a society
- C) A system that consists only of language and religion
- D) The natural environment that influences human behavior

3. Which agent of socialization has the first influence on a child?

- A) School
- B) Family
- C) Media
- D) Peers

4. Mead's theory of the Social Self includes?

- A) I and Me
- B) Id and Ego
- C) Self and Others
- D) Nature and Nurture

5. What is deviance?

- A) Following social norms
- B) Learning new skills
- C) Violating social norms
- D) Obeying laws

6. What is Sociology?

- A) Study of human society and social behavior
- B) Study of economic systems
- C) Study of ancient civilizations
- D) Study of individual psychology



7. Which of the following is a type of social interaction?

- | | |
|-------------------|----------------|
| A) Self-interest | C) Isolation |
| B) Individualism. | D) Cooperation |

8. What are the two main types of status

- | | |
|--------------------------|------------------------|
| A) Achieved and Ascribed | C) Personal and Social |
| B) Primary and Secondary | D) Fixed and Temporary |

9. When a person has multiple roles that create tension, it is called?

- | | |
|------------------|----------------------------------|
| A) Role strain | C) Role enhancement |
| B) Role learning | D) To critique the original text |

10. Founder of sociology?

- | | |
|-------------------|-----------------|
| A) Max Weber | C) August comte |
| B) Emile Durkheim | D) Karl Marx |



NATIONAL UNIVERSITY OF PAKISTAN

BS Program (Model Paper)
End-Semester Examination

Reg.No: _____

Course Code: GEN-109

Course Title: Sociology

Sign of Invigilator: _____

Time Allowed: 02 Hrs 40 minutes

Total Marks: 40

General Instructions

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

SECTION – B

Q.2 Give short answer to the following questions:

(5*2=10)

- What is structural functionalism theory?
- Define these terms:
 - Ascribed status
 - Ethnocentrism
 - Role Conflict
 - Society
- What is the formal and informal social control?
- Briefly write about formal agencies of socialisation.
- Discuss any five factors of social change.

SECTION – C

Write detailed answers to the following questions.

(10*3=30)

- Q. 3.** (a) Define group and its types? (5)
(b) How different groups form a personality support your answer with examples. (5)
- Q 4.** (a) Briefly discuss types of crime. (4)
(b) Discuss different causes of crime in Pakistan, how can it be controlled? (6)
- Q 5.** (a) What is cultural diversity? (3)
(b) What is the importance of cultural diversity in modern world? (7)

**ANSWER KEY**

QUESTION NUMBERS	ANSWER
1	C
2	B
3	B
4	A
5	C
6	A
7	D
8	A
9	A
10	C



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN**BS Program (Model Paper)
End-Semester Examination****Course Code: GEN-112****Time Allowed: 20 minutes****Course Title: Civics and Community Engagement****Total Marks: 10****Sign of Invigilator: _____****General Instructions**

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION-A**Q1. Choose the most appropriate option.****(10x1=10)**

- _____ is a common digital platform used for civic engagement.
a. Social Media b. Radio c. T. V d. Newspapers
- _____ shows the legal relationship between an individual and a state.
a. Community b. Nation c. Society d. Citizenship
- _____ is a fundamental principle of modern democratic societies.
a. Monarchy b. Oligarchy c. rule of law d. autocracy
- By reading _____ citizens can be aware of their rights.
a. History book b. constitution c. political history d. None of these
- Ethnic diversity refers to _____.
a. variety of opinion b. economics disparity c. political ideology d. differences in racial
- Legal rights are the privileges those _____.
a. Based on norms b. accepted by society c. recognized by law d. none of these
- Marches and sit-ins are the examples of _____.
a. Advocacy b. activism c. public opinion d. public discourse
- Nelson Mandela had spent _____ years in prison during his struggle.
a. 27 b. 12 c. 20 d. 15
- Martin Luther King was an _____ civil rights activist
a. African b. American c. Anglo d. none of these
- The term Apartheid means _____.
a. Ethnic cleansing b. protest c. differences d. separatene



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN

BS Program (Model Paper)

End-Semester Examination

Course Code: GEN-112

Time Allowed: 02 Hrs and 40 minutes

Course Title: Civics and Community Engagement

Total Marks: 40

Sign of Invigilator: _____

General Instructions

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

SECTION-B

Q.3 Give short answer to the following:

(5x2=10)

- What are the characteristics of community?
- Name the types of Activism?
- Describe is the concept of Cyber Ethics?
- Write any three differences between Public Opinion and Advocacy?
- Define fundamental rights?

SECTION-C

Write a detailed answer to the following

(10x3=30)

Q.3. How we can have effective community engagement?

(10)

Q.4. What was the contribution of Martin Luther King in the Civil Rights Movement?

(10)

Q.5 Discuss the impact of digital devices on citizenship and the rules for digital citizenship?

(5+5)

**Answer Key**

QUESTION NUMBER	ANSWER
1	A
2	D
3	C
4	B
5	D
6	C
7	B
8	A
9	B
10	D

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: GEN-116**Time Allowed:** 20 minutes**Course Title:** Discrete Mathematics-II**Total Marks:** 10**Sign of Invigilator:** _____**General Instructions**

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A**Q1. Choose the most appropriate option.****(10*1=10)****1. The location of an element in a list is done by**

A) Linear search

C) Binary search

B) Bubble sort

D) Insertion sort

2. In the Cashier's algorithm change of 51 cents using quarters, dimes, nickels and pennies is:

A) Five- dimes one- penny

C) Ten-nickels one-penny

B) Two-quarters one-penny
nickel

D) One-quarter two-dimes one-

3. The octal expansion of $(12345)_{10}$ isA) $(1111000011)_8$ C) $(30071)_8$ B) $(32573)_8$ D) $(3B2L)_8$ **4. If $f(0) = 3$ and $f(n + 1) = 2f(n) + 3$ then $f(2) =$**

A) 3

C) 9

B) 21

D) 20

5. Total different bits strings of length seven are:

A) 128

C) 7

B) 0

D) 49

6. If n is a composite integer, then n has a prime divisor less than equal to:A) n C) $\sqrt{n}$ B) n^2 D) n^3 **7. A full m -ary tree with i internal vertices contains**A) $n = mi + 1$ verticesC) $n = mi$ verticesB) $n = mi - 1$ verticesD) $n = m$ vertices



8. The value of $1 + \bar{1}$ is

- A) 0
B) 2
C) -1
D) 1

9. The maximum number of leaves in an m -ary tree of height h are:

- A) m
B) m^h
C) m^2
D) h^m

10. Probability when two dice are rolled, the sum of numbers on two dice is 7:

- A) $7/36$
B) $1/6$
C) $1/36$
D) $1/12$

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: GEN-116**Time Allowed:** 02 Hrs and 40 minutes**Course Title:** Discrete Mathematics-II**Total Marks:** 40**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

SECTION – B**Q.2 Give short answers to the following questions:****(5*2=10)**

- Use the bubble sort to sort 6,2,3,1,5,4 showing the list obtained at each step.
- Find the greatest common divisor of 414 and 662 using the Euclidean algorithm.
- Merge the two lists 2,3,5,6 and 1,4.
- Show that for every integer n there is a multiple of n that has only 0s and 1s in its decimal expansion.
- Is the “divides” relation on the set of positive integers transitive?

SECTION – C**Write detailed answers to the following questions.****(10*3=30)****Q.3 a)** Show that x^3 is $O(x^4)$ but that x^4 is not $O(x^3)$.**(5)****b)** Use mathematical induction to show that $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$ **(5)****Q.4 a)** Show that if n is an integer greater than 1, then n can be written as the product of primes. By using Strong Induction.**(5)****b)** How many poker hands of five cards can be dealt from a standard deck of 52 cards? How many ways are there to select 47 cards from a deck of 52 cards?**(5)****Q. 5 a)** A connected multigraph has an Euler path but not an Euler circuit if and only if it has exactly two vertices of odd degrees.**(5)****b)** Sometimes light fixtures are controlled by more than one switch. Circuits need to be designed so that flipping any one of the fixtures turns the light on when it is off and turns the light off when it is on. Design circuits that accomplish this when there are two switches and when there are three switches.**(5)**

**Answer Key**

QUESTION NUMBER	ANSWER
1	A
2	B
3	C
4	B
5	A
6	C
7	A
8	D
9	B
10	B

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: GEN-118**Time Allowed:** 20 minutes**Course Title:** Calculus and Analytical Geometry**Total Marks:** 10**Sign of Invigilator:** _____**General Instructions**

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A**Q1. Choose the most appropriate option.****(10*1=10)**1. Find Critical point of $f(x) = \frac{4x}{x^2+1}$ A) $(-1, -2)$ and $(1, 2)$ C) $(1, 2)$, $(-1, 2)$ and $(0, 0)$ B) $(2, 1)$, $(0, 1)$ and $(-2, -1)$ D) $(-1, 2)$, $(1, -2)$ 2. $\lim_{x \rightarrow 2} \frac{x^3-8}{x^2-4}$ is

A) 4

C) 0

B) 1

D) 3

3. If $f(x) = x^2 - 4x + 5$, then Range of Function is

A) All Real Numbers

C) $[4, \infty]$ B) $[5, \infty)$ D) $[1, \infty)$

4. A Tangent is a line touching circle at

A) Two points

C) One point

B) No point

D) Three points

5. A point on a smooth curve where concavity changes is known as

A) Critical point

C) Inflexion point

B) Asymptotes

D) Saddle point

6. $\int \frac{1}{1+x^2} dx$ A) $\cos^{-1} x$ C) $\sec^{-1} x$ B) $\tan^{-1} x$

D) None of these

7. The distance between the line $4x + 3y - 7$ from the origin isA) $\frac{7}{5}$ C) $\frac{5}{7}$ B) $\frac{-7}{5}$ D) $\frac{-5}{7}$ 8. $f(x) = \tan x$, then $f^{-1}(1)$ isA) $\frac{\pi}{4}$ C) $\frac{\pi}{3}$

B) Does not exist

D) None of these

9. The perpendicular distance of the origin from the plane $x - 3y + 4z = 6$

A) 6

C) $\frac{6}{\sqrt{26}}$ B) $\frac{1}{\sqrt{26}}$

D) 36



10. If a function has derivative at $x = c$ then function is

- | | |
|------------------|---------------|
| A) Discontinuous | C) Decreasing |
| B) Undefined | D) Continuous |

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: GEN-118**Course Title:** Calculus and Analytical Geometry**Time Allowed:** 02 Hrs and 40 minutes**Total Marks:** 40**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

SECTION – B**Q.2 Give short answer to the following questions:****(5*2=10)**1. Find value of k if function $f(x)$ is continuous at $x = 0$

$$f(x) = \begin{cases} \frac{x^2 - x}{2x}, & x \neq 0 \\ k, & x = 0 \end{cases}$$

2. Find Second derivative of $y = 2^x \ln \left[\frac{x^2 - 1}{x^2 + 1} \right]$.

3. Find Intervals on which function is Increasing and Interval on which function is decreasing

$$f(x) = x + \sin x \quad [0, 2\pi]$$

4. Find integration of $\int x \cos(x^2 + 1) dx$.5. Find parametric equation of straight line passing through $(5, -3, 4)$ and $(1, -2, 7)$.**SECTION – C****Write answers to the following questions.****(10*3=30)****Q.3** Evaluate the limit of following function

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$$

Q.4 Find Point of inflection and check concavity of the function. Also draw the graph of the function $x^4 - 4x^3 + 1$ **Q.5** Find point of intersection of lines given below. Also find equation of plane determined by these lines

$$l_1; x = 3t + 1, y = 2t + 2, z = 2t + 1$$

$$l_2; x = t + 4, y = 2t + 4, z = -3t + 3$$

**Answer Key**

QUESTION NUMBER	ANSWER
1	D
2	D
3	D
4	C
5	C
6	B
7	A
8	A
9	C
10	D



INTERDISCIPLINARY COURSES

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: BOT-106**Course Title:** Fundamentals of Genetics and Evolution**Time Allowed:** 20 minutes**Total Marks:** 10**Sign of Invigilator:** _____**General Instructions**

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A

Q1. Choose the most appropriate option.

(10*1=10)

- Chromosomes segregate during
 - cell division
 - chromosomal division
 - meiosis
 - mitosis
- The gene causing the white eye trait in *Drosophila* resides on the
 - X chromosome
 - Y chromosome
 - both a and b
 - none
- According to Watson and Crick the replication of DNA is
 - semi conservative
 - conservative
 - dispersive
 - none
- Genetic code is the combination of Nucleotides.
 - 3
 - 2
 - 4
 - 64
- The mating in which an individual showing a dominant phenotype is crossed with an individual showing its recessive phenotype is called
 - test cross
 - back cross
 - cross above
 - none
- Linked gene whose loci are close to each other do not obey Mendel's law of independent assortment, because these cannot assort independently during
 - meiosis
 - mitosis
 - both a and b
 - none
- Those part of the body used extensively to cope with the environment become larger and stronger, while those that are not used deteriorate. This idea was given by
 - Lynn Margulis
 - Aristotle
 - Lamarck
 - Darwin
- Hardy- weingberg theorem is used to calculate gene frequency in population in
 - Dynamic equilibrium
 - equilibrium
 - none
 - both a and b
- Evolutionary change is based mainly on the interactions between populations of organisms and their
 - Atmosphere
 - environment
 - habitat
 - all of the above
- refers to the process that have transformed life on earth from it's earliest forms to vast diversity that we observe today.
 - evolution
 - co-evolution
 - convergent evolution
 - divergent evolution

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: BOT-106**Course Title:** Fundamentals of Genetics and Evolution**Time Allowed:** 02 Hrs and 40 minutes**Total Marks:** 40**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

SECTION – B**Q.2 Give short answer to the following:****(5*2=10)**

- Define Law of segregation. Give Mendelian examples.
- What is sex determinant mechanism in drosophila?
- What are the main hypothesis about the mode of replication of DNA?
- Differentiate between founder effect and bottle effect.
- Define artificial selection. Give it's aims and objectives.

SECTION – C**Write a detailed answer to the following:****(10*3=30)****Q.3 "Gene linkages influences the Mendelian ratio" Justify with example. (10)****Q.4 Explain the mechanism of sexual transformation of genetic material in bacteria. (10)****Q.5 Explain the following**

- Hardy-Weingberg theorem **(05)**
- Role of mutation and gene flow in evolution **(05)**

**Answer Key**

QUESTION NUMBER	ANSWER
1	A
2	A
3	A
4	A
5	A
6	A
7	C
8	B
9	B
10	A

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: BOT-106L**Time Allowed:** 02 Hrs**Course Title:** Fundamentals of Genetics and Evolution-Lab**Total Marks:** 30**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

Attempt the following questions.**Q.1 Briefly answer the following.****(6+2+2=10)**

- a) Difference between 9:3:3:1 and 3:1 ratio with example.
- b) Determine the possible genotype of yellow seeded plant when it is test crossed with green seeded plants.
- c) A cross between two green corn plants yield 2/3 progeny that are green and 1/3 progeny is white. What is the genotype of the green parents' plants?

Q.2 What is the 3 point cross and how sequence of the linear gene can be determined? Also mention genetic distance of two and three genes involved.**(10)****Q.3 Viva Voce + Practical Notebook****(05+05 = 10)**



Reg.No:_____

NATIONAL UNIVERSITY OF PAKISTAN

BS Program (Model Paper)
End-Semester Examination

Course Code: MATH-111

Time Allowed: 20 minutes

Course Title: Calculus-II

Total Marks: 10

Sign of Invigilator: _____

General Instructions

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A

Q1. Choose the most appropriate option.

(10*1=10)

1. The function $y = f(x)$ is 1 – 1 if its graphs intersect each horizontal line at most

- a) once b) twice
c) thrice d) none

2. $\ln(\sin\theta) - \ln\left(\frac{\sin\theta}{5}\right)$ is equal to

- a) $\ln(\sin\theta)$
c) $\ln(\sin\theta - 5)$

3. $\int_0^{\sqrt{\ln \pi}} 2xe^{x^2} \cos(e^{x^2}) dx$ is equal to

- a) 0.84147 b) 0.94147
c) -0.84147 d) -0.94147

4. Euler solution y_1 of $y' = 1 + y$, $y(0) = 1$ and step size $dx = 0.05$ is

- a) 4.1 b) 3.1
c) 2 d) 1.1

5. The value of k, if $100e^{10k} = 200$ is

- a) $\frac{\ln 2}{10}$ b) $\frac{\ln 100}{10}$
c) $\frac{\ln 2}{100}$ d) $\frac{\ln 2}{200}$

6. The Taylor series of $\cos x$ is

- $$\begin{array}{ll} \text{a) } \sum_{k=0}^{\infty} \frac{(-1)^{k+1} x^k}{(2k+1)!} & \text{b) } \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} \\ \text{c) } \sum_{k=2}^{\infty} \frac{(-1)^{k+1} x^k}{(2k+1)!} & \text{d) } \sum_{k=2}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} \end{array}$$

7. The polar equation for the circle $x^2 + (y - 3)^2 = 9$ is

- a) $r = 9\sin\theta$
b) $r = 3\sin\theta$
c) $r = 6\sin\theta$
d) $r = 7\sin\theta$



8. The focus and directrix of the parabola $y^2 = 10x$ is

- a) $(\frac{5}{2}, 0)$ and $-\frac{5}{2}$ b) $(\frac{5}{2}, 0)$ and 0
c) $(0, \frac{5}{2})$ and $-\frac{5}{2}$ d) $(0, \frac{5}{2})$ and 0

9. $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$ is

- a) $1/3$ b) $1/4$
c) $1/5$ d) $1/6$

10. The derivative of all $y = 7^{\sec \theta} \ln 7$ with respect to θ is

- a) $7^{\sec \theta} (\ln 7) \sec \theta \tan \theta$ b) $7^{\sec \theta} (\ln 7)^2 \sec \theta \tan \theta$
c) $7^{\sec \theta} (\ln 7)^2 \sec^2 \theta$ d) $7^{\sec \theta} (\ln 7)^2 \tan^2 \theta$

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: MATH-111**Time Allowed:** 02 Hrs and 40 minutes**Course Title:** Calculus-II**Total Marks:** 40**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

SECTION – B**Q.2 Give short answer to the following questions:****(02*05=10)**

- a) Find the derivative with respect to
- θ

$$y = \ln \left(\frac{\sqrt{\sin \theta \cos \theta}}{1 + 2 \ln \theta} \right)$$

- b) Solve the first order differential equation

$$xy' = x^2 + 3y \quad x > 0 \text{ with initial condition } y(1) = 2.$$

- c) Evaluate
- $\int_1^{\infty} \frac{\ln(x) dx}{x^2}$
- .

- d) Evaluate using power series
- $\lim_{x \rightarrow 0} \frac{\sin x - \tan x}{x^3}$

- e) Find the center and radius of the sphere
- $x^2 + y^2 + z^2 + 3x - 4z + 1 = 0$

SECTION – C**Write detailed answers to the following questions.****(10*03=30)**

- Q.3. a)**
- Find an equation for the line tangent to the graph of
- $y = \cot^{-1} x$
- at
- $x = -1$
- . (05)

- b)**
- Find the inverse of the function if
- $f(x) = \frac{\sqrt{x}}{\sqrt{x-3}}$
- (05)

- Q.4. a)**
- For what values of
- p
- the integral
- $\int_1^{\infty} \frac{dx}{x^p}$
- converges and diverges? When the integral does converge, what its value? (07)

- b)**
- Investigate the converges of the series
- $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$
- . (03)

- Q.5. a)**
- The coordinate axes are to be rotated through an angle
- α
- to produce an equation for the curve
- $2x^2 + \sqrt{3}xy + y^2 - 10 = 0$
- that has no cross product term. Find
- α
- and the new equation. Identify the curve. (07)

- b)**
- Find a vector perpendicular to the plane of
- $P(1, -1, 0)$
- ;
- $Q(2, 1, -1)$
- and
- $R(-1, 1, 2)$
- . (03)

**Answer Key**

QUESTION NUMBER	ANSWER
1	A
2	B
3	C
4	D
5	A
6	B
7	C
8	A
9	D
10	B



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN**BS Program (Model Paper)
End-Semester Examination****Course Code:** POL-212**Time Allowed:** 20minutes**Course Title:** Political Science**Total Marks:** 10**Sign of Invigilator:** _____**General Instructions**

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION –A**Q1. Choose the most appropriate option.****(10x1=10)**

1. State has _____ elements.
a. 5 b. 4 c. 3 d. 6
2. The term "Politics" is derived from the _____ word.
a. Latin. b. Greek c. German d. English
3. _____ demands complete subservience to the state.
a. Democracy b. Political Party c. People d. Totalitarian state
4. Sovereignty belongs to _____.
a. State. b. Government c. Council d. Society
5. Hitler belonged to _____.
a. Italy b. Germany c. U. K d. France
6. A government has _____ branches.
a. 4 b. 2 c. 3 d. 5
7. Through _____ elections representatives are elected by public.
a. direct b. indirect c. by election d. referendum
8. The word "nation" has been derived from "natio" which means _____.
a. Family b. Society c. State d. Born
9. _____ enforces the law
a. Executive b. Legislature c. Judiciary d. none of these
10. The fundamental law of the state is called _____.
a. civil law b. constitution c. ordinance d. None of these



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN

BS Program (Model Paper)

End-Semester Examination

Course Code: POL-212

Time Allowed: 02 Hrs and 40 minutes

Course Title: Political Science

Total Marks: 40

Sign of Invigilator: _____

General Instructions

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

SECTION-B

Q.2 Give short answer to the following:

(5x2=10)

- a) What is the significance of Political Science?
- b) How does the term "Bill" & "Law" are different from each other?
- c) Describe the role of Judiciary in a state?
- d) What is meant by direct election?
- e) Write any four characteristics of power?

SECTION-C

Write a detailed answer to the following

(10x3=30)

Q.3. Explain the characteristics of democracy?

(10)

Q.4. What does legislative branch of government do in a state?

(10)

Q.5 Write the concept of Nation and describe its elements?

(3+7)

**Answer Key**

QUESTION NUMBER	ANSWER
1	B
2	B
3	D
4	A
5	B
6	C
7	A
8	D
9	A
10	B

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: STAT-102**Time Allowed: 20 minutes****Course Title: Statistics – II****Total Marks: 10****Sign of Invigilator: _____****General Instructions**

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION-A**Q1. Choose the most appropriate option.****(10x1=10)**

- If n is the number of elements of a set, the total number of subsets of this set is:
 - $2n$
 - n^2
 - 2^n
 - $n!$
- What is the probability of drawing a king of spade from a pack of 52 cards?
 - $1/11$
 - $1/13$
 - $1/26$
 - $1/52$
- What is the value of p when binomial distribution is positively skewed?
 - $p < q$
 - $p > q$
 - $p = q$
 - Both a & b
- If $S.d(X) = 2$ then $Var(4X + 8) =$
 - 16
 - 72
 - 64
 - 24
- If mean and S.D. of a population are 20 and 5 respectively. What would be the standard error of means if the sample size is 25?
 - 5
 - 1
 - 25
 - 4
- A misfit person is selected for the job is an example of:
 - Type-I error
 - Type-II error
 - No error
 - None of these
- Which test statistic should be preferred to test the population mean if population variance is known?
 - Z
 - t
 - F
 - Chi
- In a Z test degree of freedom is:
 - $n+1$
 - $n-2$
 - $n+2$
 - $n-1$
- If the attributes "A" and "B" are independent, then the coefficient of association is:
 - Negative
 - Positive
 - Zero
 - None of these
- The Wilcoxon Signed-Rank Test is used for comparing samples.
 - Independent
 - Paired
 - large
 - Categorical

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: STAT-102

Course Title: Statistics – II

Sign of Invigilator: _____

Time Allowed: 02 Hrs and 40 minutes

Total Marks: 40

General Instructions

- Do not write anything on the question paper other than Registration No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question papers.

SECTION – B**Q.2 Give short answer to the following:****(5x2=10)**

- Assume that X is a number chosen at random from the set of integers 1 and 15 inclusive. What is the probability that X is a multiple of 3?
- Write the properties of binomial experiment.
- A random sample of 500 pineapples was taken from a large consignment and 65 were found to be bad. Show that the percentage of bad pineapples in the consignment almost certainly lies between 8.5 and 17.5.
- Define: i. Sample Space ii. Sampling Frame iii. Critical Region iv. Type – I Error
- Given $\bar{x} = 120$, $\mu = 100$, $n = 25$, $t = 2.88$. Find the value of s .

SECTION – C**Write detailed answers of the following questions.****(3x10=30)**

- Q.3.** Draw all possible samples of size 3 without replacement from the following population 2, 3, 5, 9, 9, 11. Make sampling distribution of the proportion of prime numbers. Also verify the results.
(10)
- Q.4.** Out of 900 persons, 300 are literates. Among the illiterates, 400 have not travelled beyond the limits of their district. Two hundred of the literates are among those who have travelled. Find out by calculating the value of chi-square statistic if there is any association between travelling and literacy. Use 5% level of significance.
(10)
- Q.5** Explain the framework of Kruskal Wallis Test in detail.
(10)

**Answer Key**

QUESTION NUMBER	ANSWER
1	C
2	D
3	A
4	C
5	B
6	B
7	A
8	D
9	C
10	B

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: STAT-111

Time Allowed: 20 minutes

Course Title: Statistics

Total Marks: 10

Sign of Invigilator: _____

General Instructions

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A**Q1. Choose the most appropriate option.****(10x1=10)****1. Statistic is a numerical quantity which is calculated from:**

- a. Population b. Data c. Parameter d. Sample

2. If a scale has an absolute zero then it is called

- a. Ordinal b. Nominal c. Ratio d. Interval

3. The variance of a set of 50 observations is 64. If each observation is multiplied by 2, then the value of S.D. will be:

- a. 4 b. 8 c. 16 d. 32

4. Which of the following measure is most affected by extreme values?

- a. Mean b. Median c. Mode d. Range

5. If a distribution is abnormally tall and peaked then it is called:

- a. Mesokurtic b. Leptokurtic c. Platykurtic d. Flatykurtic

6. If one variable decreases with the decrease in the other variable then the correlation is:

- a. Positive b. Negative c. Increasing d. Decreasing

7. If X variable assumes the values 1, 2 and 3 and Y variable assumes the values 2, 4, and 6 then the regression coefficient Y on X is:

- a. 1 b. 2 c. 1/2 d. 4

8. Paasche's index number has a/an:

- a. Forward bias b. Upward bias c. Backward bias d. Downward bias

9. Increase in prices of a commodity due to earthquake is an example of:

- a. Regular variation b. Seasonal variation c. Irregular variation d. Secular Trend

10. If Laspeyre's index number is 110 and Fisher's index number is 128 then Paasche's index number is approximately:

- a. 120 b. 129 c. 141 d. 149



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN**BS Program (Model Paper)
End-Semester Examination****Course Code:** STAT-111**Time Allowed:** 02 Hrs and 40 minutes**Course Title:** Statistics**Total Marks:** 40**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Registration No.
- Attempt all questions. All the Question & Answered must be numbered as mentioned in the question paper.

SECTION – B**Q.2 Give short answer to the following:****(5x2=10)**

- Differentiate between histogram and historigram.
- For the values 2,4,5,5,7,8 and 9. Find lower and upper quartiles.
- The mean wage of 4 men is Rs. 170 per week. What is the mean wage of further 6 men if the mean wage of 10 men is Rs. 200?
- Define: i. Skewness ii. Kurtosis iii. Primary Data iv. Discrete Variable
- Find mode(s) and coefficient of variation for the following data of marks 3,4,5,7,3,4,8,9,3,6.

SECTION – C**Write detailed answers of the following questions.****(3x10=30)****Q.3. Given the following data****(10)**

Time (X)	1.5	2	3.5	4	5.5	6	7
Temp.(Y)	10	8	8	7	5	3	2

Find

- Regression equation Y on X.
- Show that correlation coefficient is GM of two regression coefficients.

Q.4. Find 4 year centered moving averages for the data given below:**(10)**

Years	Quarters			
	I	II	III	IV
2019	105	125	138	170
2020	109	135	142	160
2021	112	139	145	170
2022	115	144	150	175



Q.5 Find base year weighted and current year weighted price and quantity index numbers. Also find Fisher's price and quantity index numbers for the following data. **(10)**

2010(Base)		2020	
Price	Quantity	Price	Quantity
50	20	77	18
60	18	88	15
65	15	87	14
72	16	95	13
75	17	99	15

**Answer Key**

QUESTION NUMBER	ANSWER
1	C
2	D
3	A
4	C
5	B
6	B
7	A
8	D
9	C
10	B



NATIONAL UNIVERSITY OF PAKISTAN

BS Program (Model Paper)
End-Semester Examination

Reg.No: _____

Course Code: WC-201

Time Allowed: 20 minutes

Course Title: عالمی کلاسیک کے تراجم

Total Marks: 10

Sign of Invigilator: _____

ہدایات: حذف کر کے / کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ براہ کرم لیڈ پینسل کا استعمال نہ کریں۔
حصہ اول (حصہ معروضی) کے تمام اجزاء کا جواب اسی پرچے پر دیں۔

حصہ اول

(10*1=10)

سوال نمبر 1: مناسب ترین جواب کا انتخاب کریں۔

۱. بومر کس زبان کا شاعر ہے؟
(ا) عبرانی (ب) سریانی (ج) یونانی (د) سنس کِرت
۲. میکتھ، اوتھیلواور بیملٹ کس ادیب کی تخلیقات ہیں؟
(ا) جے خف (ب) شیکسپیئر (ج) دستوئیفسکی (د) مارکیز
۳. عظیم ناول نگار مارکیز کس ملک کے رہنے والے تھے؟
(ا) سپین (ب) کولمبیا (ج) پرتگال (د) سینیگال
۴. "جرم و سزا" اور "کرامازوف برادران" کس ادیب کے بہت اہم ناول ہیں؟
(ا) جے خف (ب) دستوئیفسکی (ج) شیکسپیئر (د) مارکیز
۵. "تنہائی کے سو سال" کس کا ناول ہے؟
(ا) مارکیز (ب) دستوئیفسکی (ج) شیکسپیئر (د) جے خف
۶. ایلٹ کی شاہ کار نظم "The Waste Land" کا بنیادی موضوع کیا ہے؟
(ا) تاریخ نویسی (ب) مذہب (ج) سیاست (د) دورِ جدید کا تہذیبی اور روحانی زوال
۷. "جعلی جواہرات" اور "روٹی کے گناہ" کس کہانی کار کی تخلیقات ہیں؟
(ا) جے خف (ب) دستوئیفسکی (ج) مارکیز (د) موپساں
۸. انتون جے خف عالمی ادب میں کس صنفِ ادب کی وجہ سے معروف ہیں؟
(ا) سفر (ب) خودنوشت سوانح (ج) افسانہ (د) داستان
۹. "اوڈیسی" کس شاعر کی مشہور زمانہ تصنیف ہے؟
(ا) شیکسپیئر (ب) گوٹے (ج) بومر (د) ایلٹ
۱۰. جادوئی حقیقت نگاری کس ادیب کی پہچان بنی؟
(ا) مارکیز (ب) جے خف (ج) موپساں (د) دستوئیفسکی

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: WC-201**Course Title:** عالمی کلاسیک کے تراجم**Sign of Invigilator:****Time Allowed:** 02 Hrs and 40 minutes**Total Marks:** 40

ہدایات

سوالیہ پرچے پر رول نمبر کے علاوہ کچھ نہ لکھیں۔ تمام سوالات اور جوابات کو سوالیہ پرچے کے مطابق نمبر دیں۔

سوال نمبر 2 : مندرجہ ذیل سوالات کے مختصر جوابات تحریر کیجیے۔

(5x2 = 10)

حصہ دوم

- i. "ایلیڈ" اور "اوڈیسی" کس تہذیب کی ترجمان ہیں؟
- ii. موپسان کے افسانوں کے موضوعات کی نشان دہی کیجیے۔
- iii. دستوئیفسکی کو عظیم ناول نگار کیوں مانا جاتا ہے؟
- iv. ایلیڈ کی شاعری کی دو خصوصیات کے متعلق مختصر نوٹ تحریر کیجیے۔؟
- v. مارکیز کے کسی ایک ناول پر تبصرہ کیجیے۔

حصہ سوم

(3x10=30)

سوال نمبر 3: شیکسپیئر کے کسی ایک ڈرامے کا خاصہ لکھیے۔

10

سوال نمبر 4: ایلیڈ کی تین مشہور نظموں کا تعارف لکھیے۔

10

سوال نمبر 5: انتون چخوف کی افسانہ نگاری کی اہم خصوصیات کی نشان دہی کیجیے۔

10



دُرست جوابات : معروضی حصہ

ج	1
ب	2
ب	3
ب	4
ا	5
د	6
د	7
ج	8
ج	9
ا	10



DISCIPLINARY COURSES

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: BOT-104**Time Allowed:** 20 minutes**Course Title:** Biodiversity and Conservation**Total Marks:** 10**Sign of Invigilator:** _____**General Instructions**

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A**Q1. Choose the most appropriate option.****(10*1=10)**

1. What is the term used for variety of life forms within an ecosystem?
 - a) Population
 - b) Species
 - c) Biodiversity
 - d) Community
2. Which of the following is said to be the most important cause for the extinction of animals and plants?
 - a) Loss of habitat and fragmentation
 - b) Over exploitation of species
 - c) Invasion of alien species
 - d) Co- extinctions
3. Currently how much area of Earth's surface is covered by Tropical Rain Forests?
 - a) 90%
 - b) 50%
 - c) 1%
 - d) 6%
4. An example of human activity that causes animals to go extinct is:
 - a) Over hunting
 - b) Habitat destruction
 - c) Urban development
 - d) Introduction of predators
 - e) All of above
5. Species inventory data is an important component of :
 - a) Base line data
 - b) EIA
 - c) Red Book Data
 - d) Particular Area
6. Choose an example of Ex-situ conservation.
 - a) National Parks
 - b) Animal Reserves
 - c) Cropland
 - d) Local Communities
7. The ----- convention is an international Treaty that aims to conserve wetlands and their resources.



- a) ESA
 - b) CBD
 - c) Ramsar
 - d) MBTA
8. IUCN has maintained the Red Data Book since;
- a) 1961
 - b) 1964
 - c) 1965
 - d) 1966
9. Pakistan Biodiversity Action Plan was developed in:
- a) 2000
 - b) 2001
 - c) 2002
 - d) 2003
10. Gene bank is also known as:
- a) Genetic variety
 - b) Seed bank
 - c) Conservation bank
 - d) None



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN**BS Program (Model Paper)
End-Semester Examination****Course Code:** BOT-104**Time Allowed:** 02 Hrs and 40 minutes**Course Title:** Biodiversity and Conservation**Total Marks:** 40**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

SECTION – B**Give short answer to the following:****(5*2=10)**

- Define the following terms with examples
 - Species diversity
 - Genetic diversity
- How species become extinct? State at least five causes.
- Differentiate between endangered, threatened and vulnerable species.
- Describe at least four protected areas of Pakistan for conservation of certain animal species.
- Enlist the set of criteria used by IUCN Red List to evaluate the extinction risk of species.

SECTION – C**Write a detailed answer to the following:****(10*3=30)**

Q.3: What are biodiversity hotspots? Also elaborate the tropical ecosystem and coral reef ecosystem. Describe major and current threats to biodiversity.

Q.4: Describe the theory of mass extinction. Also mention a timeline of mass extinction.

Q.5: Public awareness is essential for biodiversity conservation. Discuss some effective awareness techniques for biodiversity conservation.

**Answer Key**

QUESTION NUMBER	ANSWER
1	C
2	A
3	D
4	E
5	B
6	C
7	C
8	B
9	A
10	B



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN
BS Program (Model Paper)
End-Semester Examination

Course Code: BOT-105

Time Allowed: 20 minutes

Course Title: Fundamentals of Taxonomy

Total Marks: 10

Sign of Invigilator: _____

General Instructions

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A**Q1. Choose the most appropriate option. (10*1=10)****I. The classifications of plants are based on vascular tissues and ____.**

- a) Roots b) Epidermal tissues c) Mesophyll tissues d) Reproductive tissues

II. The original type specimen of a species which has been indicated by the author is known as ____.

- a) Holotype b) Lectotype c) Neotype d) Isotype

III. Fusiform taproot is found in:

- a) Carrot b) Turnip c) Beet d) Cuscuta

IV. The arrangement of the leaves on the stem is called ____.

- a) Venation b) Phyllotaxis c) Remal d) Sheathing

V. Glumes are found in:

- a) Sunflower b) Colocasia c) Oak d) Wheat

VI. Nodules in roots in which bacteria live are formed in family:

- a) Poaceae b) Leguminosae c) Liliaceae d) Brassicaceae

VII. Mostly inflorescence in family Euphorbiaceae is ____.

- a) Umbel b) Head c) Cyathium d) Corymb

VIII. Spiklets venation in family Poaceae is ____.

- a) Reticulate b) Parallel c) Both a & b d) None

IX. Onion belongs to family ____.

- a) Poaceae b) Liliaceae c) Brassicaceae d) Asteraceae

X. Wheat and rice belong to family ____.

- a) Poaceae b) Liliaceae c) Brassicaceae d) Asteraceae

**NATIONAL UNIVERSITY OF PAKISTAN**

Reg.No: _____

**BS Program (Model Paper)
End-Semester Examination****Course Code:** BOT-105**Course Title:** Fundamentals of Taxonomy**Sign of Invigilator:** _____**Time Allowed:** 02 Hrs and 40 minutes**Total Marks:** 40**General Instructions**

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

SECTION – B**Give short answer to the following:****(5*2=10)**

- What is modern classification system? Give its example.
- Differentiate between phylloclade and cladode.
- What is capitulum or head? Give its example.
- How will you define phyllotaxis and venation?
- Name four plants of family Fabaceae.

SECTION – C**Write a detailed answer to the following:****(10*3=30)**

- Q.3** Discuss in detail the classification system of Engler and Prantle. Also discuss merits and elements of this system.
- Q.4** What is placentation? Discuss its different types. Also draw the diagrams.
- Q.5** Discuss in detail vegetative, floral characters & economic importance of the family Solanaceae.

**Answer Key**

QUESTION NUMBER	ANSWER
1	D
2	A
3	A
4	B
5	D
6	B
7	C
8	B
9	B
10	A

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: BOT-105L**Time Allowed:** 02 Hrs**Course Title:** Fundamentals of Taxonomy-Lab**Total Marks:** 30**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper

Attempt the following questions.**Q1.** Identify the given specimen in botanical terms and write short notes on it. (05 marks)

- Apple
- Tomato

Q2. Give a detailed description of the flowers with the help of keys. Also, mention its family. (10 marks)

1. Calyx
2. Corolla
3. Androecium
4. Gynoecium
5. Floral formula

Q3. Herbarium (05 marks)**Q4.** Viva / Practical notebook (10 marks)



Reg No: _____

NATIONAL UNIVERSITY OF PAKISTAN
BS Program (Model Paper)
End-Semester Examination

Course Code: CCC-112**Time Allowed:** 20 minutes**Course Title:** Object Oriented Programming**Total Marks:** 10

Sign of Invigilator: _____

General Instructions

Deleting/Overwriting is not allowed. Do not use lead pencil.

All parts of objective section are to be answered on this paper and handed over to the invigilator.

SECTION – A

Q1. Choose the most appropriate option. (10x1=10)

i. Which feature of OOP is indicated by the following code?

```
class student{ int marks; };  
class topper : public student{ int age; topper(int age){ this.age=age; } };
```

- | | |
|----------------------------------|---------------------------------|
| a. Encapsulation and Inheritance | b. Inheritance and polymorphism |
| c. Polymorphism | d. Inheritance |

ii. How many types of access specifiers are provided in OOP (C++)?:

- | | | | |
|------|------|------|------|
| a. 4 | b. 3 | c. 2 | d. 1 |
|------|------|------|------|

iii. Which among the following can show polymorphism?

- | | | | |
|-------------------|-------------------|----------------|-------------------|
| a. Overloading && | b. Overloading << | c. Overloading | d. Overloading += |
|-------------------|-------------------|----------------|-------------------|

iv. Which among the following represents correct constructor?

- | | | | |
|-----------------|----------------|----------------|-----------------|
| a. ~classname() | b. classname() | c. ()classname | d. ~classname() |
|-----------------|----------------|----------------|-----------------|

v. Which type of members can't be accessed in derived classes of a base class?

- | | | | |
|------------------------|--------------|------------|-----------|
| a. All can be accessed | b. Protected | c. Private | d. Public |
|------------------------|--------------|------------|-----------|

vi. Which keyword is used to declare virtual functions?

- | | | | |
|---------|--------------|------------|--------------|
| a. virt | b. virtually | c. virtual | d. anonymous |
|---------|--------------|------------|--------------|

vii. Instance of which type of class can't be created?

- | | | | |
|-----------------|-------------------|--------------------|-----------------|
| a. Parent class | b. Abstract class | c. Anonymous class | d. Nested class |
|-----------------|-------------------|--------------------|-----------------|

viii. If there is an abstract method in a class then, _____

- | | |
|---------------------------------|---|
| a. Class must be abstract class | b. Class may or may not be abstract class |
| c. Class is generic | d. Class must be public |



- ix.** Protected access is same as default access that is given implicitly in java if no specifier is mentioned.
- a. True b. False c. Both d. None of these
- x.** Which feature of OOP indicates code reusability?
- a. Encapsulation b. Inheritance c. Abstraction d. Polymorphism



Reg No: _____

NATIONAL UNIVERSITY OF PAKISTAN**BS Program (Model Paper)
End-Semester Examination****Course Code:** CCC-112
Minutes**Time Allowed:** 2 Hours 40**Course Title:** Object Oriented Programming**Total Marks:** 40

Sign of Invigilator: _____

General Instructions

- Do not write anything on the question paper other than Registration No.
- Attempt all questions. All the Q&A must be numbered as mentioned in the question paper.

SECTION – B

Q.2 Give short answer to the following:

(5x2=10)

- Write a C++ program to create a class called Car that has private member variables for company, model, and year. Implement member functions to get and set these variables.
- Discuss the use of public, private and protected access specifiers and their visibility in the class.
- Discuss the role of access specifiers in inheritance and show their visibility when they are inherited as public, private and protected.
- Which operators cannot be overloaded? Write steps to overload + operator so that it can add two complex numbers. Explain with example how can a function template be created.
- What is Try, Catch and Throw? Explain with example.

SECTION – C

Write detailed answers of the following questions. (3x10=30)

Q.3 A hotel booking system manages different room types: Standard, Deluxe, and Suite.

Requirements:

- Create a **Base Class** Room with:
 - roomNumber (int)
 - pricePerNight (double)
- A virtual function calculate **Total Cost** (int nights)
- Create a Derived Class StandardRoom that calculates cost as pricePerNight * nights



- Create a Derived Class DeluxeRoom that applies a 10% surchargePa
- Create a Derived Class SuiteRoom that applies a 20% surcharge
- In main (), allow a user to select a room type, input the number of nights, and calculate the total cost.

Q.4 An airline reservation system handles different classes of tickets: **Economy, Business, and First Class.**

Requirements:

- Create a **Base Class** Ticket with:
 - passengerName (string)
 - seatNumber (int)
 - A virtual function showTicketDetails()
- Create a **Derived Class** EconomyClass that overrides showTicketDetails()
- Create a **Derived Class** BusinessClass that overrides showTicketDetails()
- Create a **Derived Class** FirstClass that overrides showTicketDetails()
- In main(), create an **array of pointers to Ticket** and use **late binding** to display ticket details dynamically.

Q.5 A company wants to secure employee salary details.

Requirements:

Create a class Employee with:

- employeeID (private)
- name (private)
- salary (private)
- Getter for salary but no setter (salary should not be modified directly)
- Function increaseSalary(double percent) to update salary

In main():

- Create an employee, increase salary, and display updated detail

**Answer Key**

Question Number	Answer
1.	A
2.	B
3.	A
4.	B
5.	C
6.	C
7.	B
8.	A
9.	B
10.	B

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: CCC-112**Time Allowed:** 02 Hrs and 40 minutes**Course Title:** Object Oriented Programming**Total Marks:** 40**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Registration No.
- Attempt all questions. All the Q&A must be numbered as mentioned in the question paper.

SECTION – B**Q.2 Give short answer to the following:****(5x2=10)**

- Write a C++ program to create a class called Car that has private member variables for company, model, and year. Implement member functions to get and set these variables.
- Discuss the use of public, private and protected access specifiers and their visibility in the class.
- Discuss the role of access specifiers in inheritance and show their visibility when they are inherited as public, private and protected.
- Which operators cannot be overloaded? Write steps to overload + operator so that it can add two complex numbers. Explain with example how can a function template be created.
- What is Try, Catch and Throw? Explain with example.

SECTION – C**Write detailed answers of the following questions.****(3x10=30)****Q.3** A hotel booking system manages different room types: Standard, Deluxe, and Suite.**Requirements:**

- Create a **Base Class** Room with:
 - roomNumber (int)
 - pricePerNight (double)
- A virtual function **calculateTotalCost** (int nights)
- Create a Derived Class Standard Room that calculates cost as pricePerNight * nights
- Create a Derived Class Deluxe Room that applies a 10% surcharge
- Create a Derived Class Suite Room that applies a 20% surcharge
- In main(), allow a user to select a room type, input the number of nights, and calculate the total cost.

Q.4 An airline reservation system handles different classes of tickets: **Economy, Business, and First Class.**

**Requirements:**

- Create a **Base Class** Ticket with:
 - passengerName (string)
 - seatNumber (int)
 - A virtual function showTicketDetails()
- Create a **Derived Class** EconomyClass that overrides showTicketDetails()
- Create a **Derived Class** BusinessClass that overrides showTicketDetails()
- Create a **Derived Class** FirstClass that overrides showTicketDetails()
- In main(), create an **array of pointers to Ticket** and use **late binding** to display ticket details dynamically.

Q.5 A company wants to secure employee salary details.

Requirements:

Create a class Employee with:

- employeeID (private)
- name (private)
- salary (private)
- Getter for salary but no setter (salary should not be modified directly)
- Function increaseSalary(double percent) to update salary

In main():

- Create an employee, increase salary, and display updated detail

**Answer Key**

QUESTION NUMBER	ANSWER
1	A
2	B
3	A
4	B
5	C
6	C
7	B
8	A
9	B
10	B

**NATIONAL UNIVERSITY OF PAKISTAN**

Reg.No: _____

**BS Program (Model Paper)
End-Semester Examination**

Course Code: CCC-112L

Time Allowed: 02 Hrs

Course Title: Object Oriented Programming -Lab

Total Marks: 30

Sign of Invigilator: _____

General Instructions

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper

Attempt the following questions.**Q.1** A vehicle rental system needs to store information about **Cars** and **Bikes**.**Requirements:**

- Create a **Class** `Vehicle` with attributes:
 - `brand` (string)
 - `model` (string)
- Create a **Class** `Rentable` with attributes:
 - `pricePerDay` (double)
 - A function `calculateRent(int days)`
- Create a **Derived Class** `Car`, which **inherits from both `Vehicle` and `Rentable`**.
- In `main()`, create an object of `Car`, set values, and calculate the rent for **5 days**.

Q.2 A company wants to manage its employees' information. There are two types of employees: **Permanent Employees** and **Contract Employees**.**Requirements:**

- Create a **Base Class** `Employee` with the following attributes:
 - `name` (string)
 - `id` (int)
 - `displayInfo()` (a function to display general employee details)
- Create a **Derived Class** `PermanentEmployee` with an additional attribute:
 - `salary` (double)
 - Override `displayInfo()` to show salary
- Create another **Derived Class** `ContractEmployee` with an additional attribute:
 - `hourlyRate` (double)
 - Override `displayInfo()` to show hourly rate
- In `main()`, create objects of both derived classes and call `displayInfo()`.

Viva Voce + Practical Notebook**(10)**

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: CCC-131**Time Allowed:** 20 minutes**Course Title:** Digital Logic Design**Total Marks:** 10**Sign of Invigilator:** _____**General Instructions**

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A**Q1. Choose the most appropriate option. (10x1=10)****1. Which of the following is a valid binary number?**

- a) 101012 b) 110011 c) 20301 d) 110210

2. What is the result of the logical operation A AND B if A = 1 and B = 0?

- a) 1 b) 0 c) Undefined d) Error

3. Which Boolean algebra law is applied to simplify the expression: $A+A'=1$ $A + A' = 1$ $A+A'=1$?

- a) Complement Law b) Idempotent Law c) Distributive Law d) Associative Law

4. Which of the following is NOT a type of flip-flop?

- a) SR Flip-Flop b) D Flip-Flop c) JK Flip-Flop d) XOR Flip-Flop

5. In a 4-bit binary adder, how many sum bits are produced?

- a) 1 b) 2 c) 4 d) 5

6. Which method is used for simplifying Boolean expressions with four variables?

- a) Boolean Algebra c) McCluskey
b) (KMap) d) Both b and

7. Which of the following is a characteristic of synchronous counters?

- a) All flip-flops change state simultaneously b) Flip-flops change state sequentially c) Uses external clock pulses for triggering d) None of the above

8. What is the main function of a shift register?

- a) Store data for future use b) Perform arithmetic operations c) Shift binary data left or right d) Decode binary information

9. In a 2-to-4-line decoder, how many output lines are activated?

- a) 1 b) 2 c) 3 d) 4

10. Which of the following circuits is used for the subtraction of two binary numbers?

- a) Half Adder b) Full Adder c) Subtractor d) Multiplexer

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: CCC-131**Time Allowed:** 02 Hrs and 40 minutes**Course Title:** Digital Logic Design**Total Marks:** 40**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Registration No.
- Attempt all questions. All the Q&A must be numbered as mentioned in the question paper.

SECTION – B**Q.2 Give short answer to the following:****(5x2=10)**

- Explain the difference between the NAND and NOR gates in terms of their truth tables and properties.
- Simplify the Boolean expression $A'BC + AB + ACA'BC + AB + ACA'BC + AB + AC$ using Boolean algebra.
- What is the Quine-McCluskey method? Explain its use in simplifying Boolean functions.
- Describe the operation of a 4-bit binary subtractor. How is it different from an adder?
- List and explain the differences between synchronous and asynchronous counters.

SECTION – C**Write detailed answers of the following questions.****(3x10=30)****Q.3.** Using the Full Adder concept, perform the following binary operations:**(a) Binary Addition:** $A=1011A = 1011A=1011$ and $B=1101B = 1101B=1101$ **(b) Binary Subtraction (using 2's complement method):** $A=1011A = 1011A=1011$ and $B=0110B = 0110B=0110$

Show the step-by-step process, including carry propagation for addition and the 2's complement calculation for subtraction.

Q.4. Explain the different types of shift registers (Serial-In-Serial-Out, Serial-In-Parallel-Out, Parallel-In-Serial-Out, and Parallel-In-Parallel-Out). Design a 4-bit shift register with parallel input and parallel output, explain its working, and discuss its applications in digital systems and memory storage.**Q.5** Design a 4-bit binary up counter using JK flip-flops, derive the Boolean expressions, and explain its working with a state transition table.

**Answer Key**

QUESTION NUMBER	ANSWER
1	B
2	B
3	A
4	D
5	C
6	D
7	A
8	C
9	A
10	C

**NATIONAL UNIVERSITY OF PAKISTAN**

Reg.No: _____

BS Program (Model Paper)**End-Semester Examination****Course Code:** CCC-131L**Time Allowed:** 02 Hrs and 40 minutes**Course Title:** Digital Logic Design - Lab**Total Marks:** 30**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the Q&A must be numbered as mentioned in the question paper

Attempt the following questions.**Q1. Design and Implement Combinational and Arithmetic Circuits**

- a) Design and implement a 4-to-1 Multiplexer using AND, OR, and NOT gates. Provide the truth table, derive the Boolean expression, and explain how the select lines control the output. (5)
- b) Construct a Full Adder circuit by combining two Half Adders and an OR gate. Draw the circuit diagram, derive the Sum and Carry Boolean expressions, and explain how the circuit performs binary addition. (5)

Q2. Design and Analysis of Sequential Circuits

- a) Implement a JK Flip-Flop using SR Flip-Flops and basic logic gates. Provide the excitation table, explain the working mechanism, and describe how the toggle condition is achieved. (5)
- b) Design a 3-bit Synchronous Up Counter using T Flip-Flops. Draw the circuit diagram, explain its working process, and describe the binary counting sequence from 000 to 111. (5)

Q.3 Viva Voce + Practical Notebook

(10)

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: CCC-161**Time Allowed:** 20 minutes**Course Title:** Database Systems**Total Marks:** 10**Sign of Invigilator:** _____**General Instructions**

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A**Q1. Choose the most appropriate option.****(10x1=10)****1. Which DBMS component is primarily responsible for maintaining data consistency during concurrent transactions?**

(A) Data Dictionary (B) Transaction Manager (C) Query Processor (D) Data Model

2. In the relational model, which key is used to uniquely identify each record within a relation?

(A) Foreign (B) Candidate (C) Composite (D) Alternate

3. Which normal form eliminates partial dependencies by ensuring that non-key attributes are fully functionally dependent on the primary key?

(A) 1NF (B) 2NF (C) 3NF (D) BCNF

4. In relational algebra, what is produced when performing a Cartesian product operation between two relations?

(A) Intersection (B) All pairs (C) Unique rows (D) Common rows

5. Which SQL command is used to permanently delete an entire database schema along with its contents?

(A) DELETE (B) DROP (C) REMOVE (D) ERASE

6. Which type of join retrieves all records from both tables, including unmatched rows by padding them with values?

(A) INNER (B) LEFT (C) RIGHT (D) FULL

7. What is the primary benefit of using views in a relational database environment?

(A) Speed (B) Storage (C) Security (D) Accuracy

8. Which Data Control Language (DCL) command is used to assign specific access privileges to database users?

(A) UPDATE (B) GRANT (C) DELETE (D) SELECT

9. In database transactions, what does a dirty read refer to?

(A) Uncommitted (B) Duplicate (C) Data loss (D) Bad index

10. Which NoSQL database model is best suited for representing hierarchical relationships?

(A) Document (B) Column (C) Graph (D) Key-value

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: CCC-161**Time Allowed:** 02 Hrs and 40 minutes**Course Title:** Database Management System**Total Marks:** 40**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Registration No.
- Attempt all questions. All the Q&A must be numbered as mentioned in the question paper.

SECTION – B**Q.2 Give short answer to the following:****(5x2=10)**

- Differentiate between **Logical Data Independence** and **Physical Data**, incorporating real-world examples to illustrate their implications on database management.
- Analyze the concept of Referential Integrity** within relational databases.
- Describe the **BCNF (Boyce-Codd Normal Form)** and provide a relation that violates 3NF but satisfies BCNF.
- What are the different types of **SQL Joins**? Explain **Self-Join** with a suitable SQL query.
- What is **ACID** in database transactions? Explain how these properties maintain database consistency.

SECTION – C**Write detailed answers of the following questions.****(3x10=30)****Q.3** Decompose the following **Student Records** relation into 3rd Normal Form (3NF), ensuring the preservation of all functional dependencies.

StudentID	Name	Department	Course	Instructor	InstructorPhone	Marks
101	Alice	Computer Sci	Database	Dr. Smith	123-456-7890	85
102	Bob	Computer Sci	Database	Dr. Smith	123-456-7890	78
103	Charlie	Physics	Quantum Physics	Dr. Johnson	987-654-3210	92
104	David	Mathematics	Calculus	Dr. Brown	555-123-4567	88
105	Eve	Computer Sci	Algorithms	Dr. Smith	123-456-7890	81

Q 4. Draw an ER diagram for: A University contains many Faculties. The faculties in turn are divided into several Schools. Each School offers numerous programs and each program contains many courses. Lecturers can teach many different courses and even the same course numerous times. Courses can also be taught by many lecturers. A student is enrolled in only one program but a program can contain many students. Students can be enrolled in many courses at the same time and the courses have many students enrolled.

Q5. Examine the three-level schema architecture (internal, conceptual, and external) and discuss how each layer contributes to data abstraction and data independence.

**Answer Key**

QUESTION NUMBER	ANSWER
1	B
2	B
3	B
4	B
5	B
6	D
7	C
8	B
9	A
10	C

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)****End-Semester Examination**

Reg.No: _____

Course Code: CCC-161L**Course Title:** Database Systems - Lab**Sign of Invigilator:** _____**Time Allowed:** 02 Hrs**Total Marks:** 30**General Instructions**

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper

Attempt the following questions.**Q.1. "Orders" and "Customers" Tables are given below:**

Orders Table:

OrderID	CustomerID	TotalAmount
501	C001	500
502	C002	1200
503	C001	900
504	C003	1500

Customers Table:

CustomerID	Name
C001	John Doe
C002	Jane Smith
C003	Mike Johnson

- a) Write an SQL query using a JOIN to display the customer names and their order amounts where the order amount exceeds 1000. (5)
- b) Retrieve the total amount spent by each customer using a JOIN, grouped by customer name. (5)

Q.2 "Employee" Table is given below:

Consider the Employee table:

EmpID	Name	Department	Salary
101	Alice	HR	60000
102	Bob	IT	75000
103	Charlie	Finance	85000
104	David	IT	72000
105	Eve	Marketing	68000

a) Write SQL queries to:

1. **Update** the salary of IT employees by **5%**. (3)
2. **Delete** all employees earning **below 65000**. (2)



- b) Write a query to **select** the names of employees working in the **HR** and **Finance** departments. (5)

Q.3 Viva Voce + Practical Notebook. (10)

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)****End-Semester Examination**

Reg.No: _____

Course Code: ECO-103**Time Allowed:** 20 minutes**Course Title:** Intermediate Microeconomics**Total Marks:** 10**Sign of Invigilator:** _____**General Instructions**

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A**Q1. Choose the most appropriate option.****(10*1=10)**

- If bundle A is preferred to bundle B, and bundle B is preferred to bundle C, then bundle A must be preferred to bundle C. This condition refers to axiom of:
 - Reflexiveness
 - Non-satiation
 - Transitivity
 - Completeness
- In case of perfect substitute goods, the indifference curve will be:
 - Negatively sloped straight line
 - Positively sloped straight line
 - Convex to origin
 - Concave to origin
- The curve that traces the optimum consumption bundles of a consumer when price of one good change while the price of other good and consumer income remain same is:
 - Income consumption curve (ICC)
 - Price consumption curve (PCC)
 - Income offer curve
 - Engle Curve
- If the income elasticity of demand for a good is - 0.8, the good is:
 - Superior
 - Normal
 - Giffen
 - Inferior
- The slope of total cost curve at a given level of output represents:
 - Average cost (AC)
 - Marginal cost (MC)
 - Average fixed cost (AFC)
 - Average variable cost (AVC)
- All points lying on the same isoquant curve show:
 - Same level of utility
 - Same level of profit
 - Same level of output
 - Same level of income
- The necessary condition for profit maximization is:
 - MR = MC
 - MR > AR
 - MR = P
 - Slope of MC > Slope of MR
- At the shut-down point, a firm faces loss equal to:
 - Average fixed cost
 - Average variable cost
 - Total variable cost
 - Total fixed cost
- A single firm represents the entire market under:
 - Perfect competition
 - Monopoly



- | | | |
|----|---|-----------------------------|
| | c) Oligopoly | d) Monopolistic competition |
| x. | A price-discriminating firm charges higher a price in the market where the demand for its product is: | |
| | a) Perfectly elastic | b) Unitary elastic |
| | c) More elastic | d) Less elastic |

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)****End-Semester Examination**

Reg.No: _____

Course Code: ECO-103**Time Allowed:** 02 Hrs and 40 minutes**Course Title:** Intermediate Microeconomics**Total Marks:** 40**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

SECTION – B**Q.2 Give short answer of the following questions:****(5*2=10)**

1. What is diminishing marginal rate of substitution? Show it graphically.
2. What do you mean by the weak axiom of revealed preference (WARP). Illustrate its violation in a diagram.
3. Define and graphically illustrate expansion path of a firm.
4. What is an iso-profit curve of a firm?
5. Briefly explain the second-degree of price discrimination.

SECTION – C**Write detailed answers to the following questions.****(10*3=30)****Q.3 a) What is income consumption curve (ICC)? Derive it in case of an inferior good. (05)****b) Using Slutsky substitution effect, prove that $P.E = S.E + I.E$ in the case of a Giffen good. (05)****Q.4 a) How does a firm achieve optimum combination of factors of production when it maximizes output subject cost constraint? Explain it diagrammatically. (05)****b) Using iso-profit and production curves, explain firm's output-input choices in the short run. (05)****Q.5 Explain different possibilities of short-run equilibrium of a firm under monopolistic competition. (10)**

**Answer Key**

QUESTION NUMBER	ANSWERS
1	C
2	A
3	B
4	D
5	B
6	A
7	C
8	D
9	B
10	D

**NATIONAL UNIVERSITY OF PAKISTAN**

BS Program (Model Paper)
End-Semester Examination

Reg. No: _____

Course Code: ECO-104

Time Allowed: 20 minutes

Course Title: Mathematical Economics I

Total Marks: 10

Sign of Invigilator: _____

General Instructions

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A**Q1. Choose the most appropriate option.****(10*1=10)**

- Set of all possible values that a dependent variable can adopt is called**
 - Domain.
 - Range.
 - Subset.
 - Universal set.
- Graph of following function is always a straight line**
 - Quadratic.
 - Cubic. .
 - Linear.
 - Logarithmic.
- The variable whose solution values are found from the model is called**
 - Endogenous variable.
 - Exogenous variable
 - Parameter.
 - Dependent
- Which of the following statement is true**
 - $(A-1)^{-1} = A^{-1}$.
 - $(A-1)^{-1} = A$.
 - $(A-1)^{-1} = A'$.
 - $(A-1)^{-1} = A^{-1}$.
- By interchanging rows or columns of a square matrix, Its Determinant will be**
 - Zero
 - Same with opposite sign
 - One
 - None of above.
- Rate of change of a function is found by**
 - Limit of a function.
 - Continuity. .
 - Derivative.
 - None of above.
- Partial derivative can be found from a function which is**
 - Single variable.
 - Constant.
 - Multivariate.
 - All of above. .
- According to Young's theorem cross partial derivatives are always**
 - Zero.
 - One.
 - Negative.
 - Equal.
- $f'(1)$ of the function $y=f(x)=cX^2$ is**
 - 2C.
 - 3.
 - 3C.
 - $3CX^2$.
- MR curve under monopoly has _____ the slope of AR**
 - Twice.
 - Thrice.
 - Same.
 - None of the above.

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)****End-Semester Examination**

Reg.No: _____

Course Code: ECO-104**Time Allowed:** 02 Hrs and 40 minutes**Course Title:** Mathematical Economics**Total Marks:** 40**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

SECTION – B**Q.2 Give short answer of the following questions:****(5*2=10)**

a. Explain the difference between constant and parameters.

b. Given $S_1=\{2,4,6\}$, $S_2=\{7,2,6\}$ and $S_3=\{4,2,6\}$ findi) $S_1 \cup S_2$ ii) $S_1 \cap S_2$ iii) $S_1 \cap S_2 \cap S_3$

c. Find Equilibrium solution of the following model

$$Q_d=3-P_2, Q_s=6P-4$$

d. Given $A=\begin{bmatrix} 0 & 4 \\ 1 & 3 \end{bmatrix}$ $B=\begin{bmatrix} 3 & 8 \\ 0 & 1 \end{bmatrix}$

Prove that $(A+B)' = A' + B'$

e. Solve the following inequalities

i) $3X-1 < 7x+2$

ii) $2X-1 < 6X+5$.

SECTION – C**Write detailed answers to the following questions.****(10*3=30)****Q.3. a)** Given the total cost function $C=Q^3-5Q^2+12Q+75$

Write the variable cost function. Also find derivative of total cost function by giving its economic interpretation.

b) Differentiate the following functions by using appropriate formula

i) $(3x+10)(6x^2-7x)$ ii) $(ax^2+b)(Cx+d)$

Q.4. Given the following national income model $Y=C+I+G$

$$C=200+0.75(1-t)Y$$

$$t=0.20$$

$$I=500$$

$$G=500$$

i). Find equilibrium value of Y and C.

ii). Prove that at equilibrium values $Y=C+I+G$.**Q.5.** Solve the following system of equations by matrix approach

$$4X_1 + X_2 - 5X_3 = 8$$

$$-2X_1 + 3X_2 + X_3 = 12$$

$$3X_1 - X_2 + 4X_3 = 5$$

**Answer Key**

QUESTION NUMBER	ANSWERS
1	B
2	C
3	A
4	C
5	B
6	C
7	C
8	D
9	A
10	A



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN**BS Program (Model Paper)
End-Semester Examination****Course Code:** ENG-112**Time Allowed:** 20 Minutes**Course Title:** Literary Forms and Movements**Total Marks:** 10**Sign of Invigilator:** _____**General Instructions**

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of the objective section will be answered on this paper and be handed in to the invigilator.

SECTION – A**Q1. Choose the most appropriate option.****(10x1=10)****1. Which of the following best defines the term ‘literary movement’?**

- a. A group of writers who share the same nationality
- b. A period in literature characterized by shared themes, styles, and philosophical beliefs
- c. A trend in publishing where books of a particular genre become popular
- d. A method of categorizing books in a library

2. Which statement best describes a short story?

- a. A lengthy narrative with subplots
- b. A detailed historical account with factual accuracy
- c. A brief narrative, a slice of life
- d. Loosely connected events without a central theme

3. Which literary movement emphasized reason, order, and adherence to classical principles, often reacting against the excesses of the Renaissance?

- a. Romanticism
- b. Surrealism
- c. Neoclassicism
- d. Absurdism

4. Which of the following is NOT a characteristic of Romanticism?

- a. Emphasis on emotion and individualism
- b. Reverence of nature
- c. Focus on industrial progress and rationality
- d. Fascination with the supernatural



5. **Who is the key figure of Transcendentalism?**
 - a. John Keats
 - b. Ralph Waldo Emerson
 - c. Oscar Wilde
 - d. Albert Camus
6. **Victorian literature is often associated with?**
 - a. Idealization of rural life and nature
 - b. Pessimism about industrialization and social reform themes
 - c. Emphasis on individual expression and rebellion
 - d. Celebration of irrationality and chaos
7. **Theatre of Absurd is an outcome of Existentialism. Who are the chief literary figures of Absurdism?**
 - a. Beckett, Kafka, Pinter
 - b. Raymond Williams, Walter Abish
 - c. Charles Olson, Denise Levertov
 - d. Anton Chekhov, Leo Tolstoy
8. **In which literary movement did writers like Virginia Woolf and James Joyce experiment with stream-of-consciousness narration?**
 - a. Postmodernism
 - b. Romanticism
 - c. Modernism
 - d. Realism
9. **Which of the following themes is central to Existentialist literature?**
 - a. The inevitability of progress and human rationality
 - b. The struggle to find meaning in an indifferent or chaotic world
 - c. The glorification of religious dogma and absolute truth
 - d. The idea that fate and destiny determine human actions
10. **What is the pattern of iambic pentameter?**
 - a. Five feet, stressed-unstressed
 - b. Four feet, unstressed-stressed
 - c. Five feet, unstressed-stressed
 - d. Six feet, stressed-unstressed



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN**BS Program (Model Paper)****End-Semester Examination****Course Code:** ENG-112**Time Allowed:** 02 Hrs and 40 minutes**Course Title:** Literary Forms and Movements**Total Marks:** 40**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Registration No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

SECTION – B**Q.2 Give short answers to the following:****(5x2=10)**

- a. How does Neoclassicism differ from Romanticism?
- b. What role did the Enlightenment movement play in shaping literature?
- c. In what ways does Postcolonial literature challenge the narratives created by colonial literature?
- d. How is poetry defined as a literary form?
- b. What are the core beliefs of Transcendentalism?

SECTION – C**Write detailed answers to the following.****(10x3=30)**

Q.3 (a) What are the key principles of Greek Idealism in Plato's philosophy, and how have they influenced literary thought? **(05)**

(b) What are the defining characteristics of Short Story as a literary genre? Substantiate your argument with reference to at least one short story that you have read. **(05)**

Q.4 What is the significance of Existentialism in modern literature, and how does it relate to Absurdism? **(10)**

Q.5 What is the Feminist Movement, and how has it influenced literature and social change over time? **(10)**

**Answer Key**

QUESTION NUMBER	ANSWERS
1	B
2	C
3	C
4	C
5	B
6	B
7	A
8	C
9	B
10	C

**NATIONAL UNIVERSITY OF PAKISTAN**

BS Program (Model Paper)

End-Semester Examination

Reg.No: _____

Course Code: ENG-122

Time Allowed: 20 minutes

Course Title: Introduction to Phonetics & Phonology

Total Marks: 10

Sign of Invigilator: _____

General Instructions

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and be handed in to the invigilator.

SECTION – A

- A unit of phonemes with or without consonants is a**
 - syllable
 - vowel
 - semi-vowel
 - phone
- A syllable, at maximum, can contain..... vowel(s) in it.**
 - two
 - one
 - three
 - four
- There are pure vowels in English.**
 - 44
 - 20
 - 24
 - 12
- /j/ is a**
 - vowel
 - syllabic vowel
 - semi-vowel
 - syllabic semi-vowel
- There are syllabic consonants in English.**
 - 3
 - 4
 - 5
 - 6
- [p^h] h above /p/ is a**
 - diacritic
 - suprasegmental
 - syllabic consonant
 - Non-syllabic vowel
- The process of not pronouncing a sound segment that might be present in the deliberately careful pronunciation of a word in isolation is described as a/an.....**
 - assimilation
 - elision
 - sloppiness
 - laziness
- /əb'dʒekt/, the small vertical line in transcription is a suprasegmental and it shows**
 - stress
 - primary stress
 - secondary stress
 - None
- There are diphthongs in English.**
 - 12
 - 24
 - 8
 - 20
- Versions of a phoneme are called**
 - homophones
 - allophones
 - homonyms
 - hyponyms



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN**BS Program (Model Paper)****End-Semester Examination****Course Code:** ENG-122**Time Allowed:** 2 Hrs and 40 minutes**Course Title:** Introduction to Phonetics & Phonology**Total Marks:** 40**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Registration No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

SECTION – B**Q.2 Do as directed:****(5x2=10)**

- What is the difference between Phonetics and Phonology?
- How is Phoneme related to Allophone?
- Why is IPA necessary?
- How is rhotic accent different from non-rhotic?
- How are Obstruents different from Sonorants?

SECTION – C**Write a detailed answer to the following:****(10x3=30)**

Q.3. a. A girl stops calling her father 'Papa' (instead she started calling him 'Daddy') as she steps into her teenage. When her surprised father enquired as to why, she retorted, "Come on Daddy, it ruins my lipstick!" How come?

(05)

b. A general distinction can be made among languages depending on their basic rhythm, whether they have syllable-timing or stress-timing. How are these two types of rhythm distinguished and which type characterizes the pronunciation of English and Urdu?

(05)

Q.4. Compare and contrast phonetic and phonological approaches to Syllable.

(10)

Q.5. Mostly our talk is fast and spontaneous, and it requires our articulators to move from one sound to the next without stopping and it leads to coarticulation. What are the effects of coarticulation?

(10)

**Answer Key**

QUESTION NUMBER	ANSWERS
1	A
2	B
3	D
4	C
5	B
6	A
7	B
8	B
9	C
10	B

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: FA-103**Time Allowed:** 20 minutes**Course Title:** History of Arts-I**Total Marks:** 10**Sign of Invigilator:** _____**General Instructions**

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A**Q1. Choose the most appropriate option.****(10x1=10)**

- The best-preserved group of works of art from Greece consists of:**
 - Domestic Architecture
 - Temples and Theatre
 - Mural and panel painting
 - Painting sculpture of the Archaic period
- Greek Architecture was**
 - Post and lentil system
 - Vault system
 - Dome system
 - Pillar system
- Tusan and Composite order were introduced by:**
 - Greek
 - Roman
 - Arryian
 - Egyptian
- The mural Paintings in villas around Mount Vesuvius Were a Part of:**
 - Religious rituals
 - Status Symbols
 - Interior Decorations
 - History Records
- What was the purpose of pre-historic cave painting?**
 - Art of sake
 - Superstitious belief
 - To gain magical control over hunting
 - protection from the evil forces
- The great pyramids at Gizeh were used as**
 - Mastabas
 - Burial chamber for the pharaohs
 - Treasure houses
 - Gymnasium
- What was the material used by the cave painters**
 - Egg tempera on wood panel
 - Earth colors and bristle brush
 - oil painting on stone
 - watercolors on clay



- viii. **Dipylon vases were used as:**
- | | |
|----------------------|--------------------|
| a. Flower containers | b. drinking cup |
| c. perfume flash | d. Grave monuments |
- ix. **All the animals of Ishtar Gate are called:**
- | | |
|--------------------|------------------|
| a. Over life-size | c. life-size |
| b. Below life size | d. None of these |
- x. **The architecture of Assyrian palaces has a prominent character.**
- | | |
|------------------|-----------------|
| a. Corbett vault | c. Winged being |
| b. Hieroglyphs | d. Cuniform |



NATIONAL UNIVERSITY OF PAKISTAN

**BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: FA-103

Time Allowed: 02 Hrs and 40 minutes

Course Title: History of Arts-I

Total Marks: 40

Sign of Invigilator: _____

General Instructions

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – B

Q.2 Give a short answer to the following:

(5x2=10)

- Explain Ziggurats of Ur.
- What were the three basic principles of Egyptian paintings?
- Write down the decoration of Ishtar gate.
- Draw and label doric order.
- Explain the treatment of animals in cave art.

SECTION – C

Write a detailed answer to the following:

(3x10=30)

Q.3 Roman people did relief work for the building and wall decoration. Explain.

Q.4 Roman people were an organized nation. Explain.

Q.5 Briefly explain Greek pottery.



NATIONAL UNIVERSITY OF PAKISTAN

**BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: FA-104

Time Allowed: 4 Hrs

Course Title: Fine Arts Studio-II (Painting Portraiture)

Total Marks: 50

Sign of Invigilator: _____

General Instructions

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

Q.1 Paint a composition using the given fruits/still life objects with oil colors, and follow the light source.



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN**BS Program (Model Paper)
End-Semester Examination****Course Code: HPE-111****Time Allowed: 20 minutes****Course Title: Games-II (Tennis, Volleyball, Hockey)****Total Marks: 10****Sign of Invigilator: _____****General Instructions**

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION- A**Q1. Choose the most appropriate option.****(10x1=10)**

1. Which of the following is a type of tennis stroke?
 - A) Forehand
 - B) Backhand
 - C) Serve
 - D) All of the above
2. What is the distance from the baseline to the service line on a tennis court?
 - A) 18 feet (5.49 meters)
 - B) 21 feet (6.4 meters)
 - C) 24 feet (7.32 meters)
 - D) 27 feet (8.23 meters)
3. What is the term for when a player wins a point while their opponent is serving?
 - A) Break
 - B) Hold
 - C) Ace
 - D) Fault
4. Who is credited with developing the first official rules and regulations for the game of Volleyball?
 - A) William G. Morgan
 - B) Alfred Halstead
 - C) James Naismith
 - D) Spalding
5. What is the distance from the net to the attack line?
 - A) 3 meters (9.84 feet)
 - B) 3.05 meters (10 feet)
 - C) 3.5 meters (11.48 feet)
 - D) 4 meters (13.12 feet)
6. What is the length of a standard volleyball court?
 - A) 18 meters
 - B) 20 meters
 - C) 18.29 meters (60 feet)
 - D) 20.12 meters (66 feet)
7. Where did the modern game of hockey originate?
 - A) England



- B) Canada
- C) United States
- D) Australia

8. What is the duration of a standard hockey match?

- A) 60 minutes
- B) 70 minutes
- C) 80 minutes
- D) 90 minutes

9. What is the distance from the goal line to the penalty spot?

- A) 10 meters
- B) 15 meters
- C) 20 meters
- D) 23 meters

10. What is the role of the goalkeeper in hockey?

- A) To score goals
- B) To defend the goal
- C) To take penalty corners
- D) To take free hits



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN**BS Program (Model Paper)
End-Semester Examination****Course Code:** HPE-111**Time Allowed:** 02 Hrs and 40 minutes**Course Title:** Games-II (Tennis, Volleyball, Hockey)**Total Marks:** 40**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Registration No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

SECTION – B**Q.2 Give short answer to the following:****(5x2=10)**

- Write the aims of games and sports?
- What is challenging systems?
- Write the history of tennis game?
- Write the rules of Volleyball game?
- What do you meant by penalty stroke in hockey game?

SECTION – C**Write a detailed answer to the following:****(10x3=30)****Q.3.** Write in detail the theories of Game and sports?**Q.4.** What do you know about the scoring system in Volleyball also draw the volleyball court dimensions?**Q.5.** Write in detail the rules and techniques of Hockey game?

**Answer Key**

QUESTION NUMBER	ANSWER
1	D
2	B
3	A
4	B
5	A
6	C
7	B
8	A
9	D
10	B



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN**BS Program (Model Paper)
End-Semester Examination****Course Code:** HPE-111L**Time Allowed:** 02 Hrs**Course Title:** Games-II (Tennis, Volleyball, Hockey) Lab**Total Marks:** 30**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Registration No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

SECTION – B**Q.1. Give short answer to the following:****(5x4=20)**

- a) Draw the Hockey grounds with complete dimensions?
- b) Draw the Volleyball court with complete dimensions?
- c) Demonstrate dribbling in hockey?
- d) Demonstrate the service techniques in Volleyball?

SECTION – C**Answer to the following:****(5x2=10)****Q.2.** Practical notebook.**Q.3.** Viva Voce.



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN**BS Program (Model Paper)****End-Semester Examination****Course Code: HPE-112****Time Allowed: 20 minutes****Course Title: Athletics-II (Long Races)****Total Marks: 10****Sign of Invigilator: _____****General Instructions**

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A**Q1. Choose the most appropriate option.****(10x1=10)****i. The race in which one foot must be in touch with the ground is _____**

- a) Running
- b) Jogging
- c) Both A & B
- d) Walking

ii. _____ is the organized program for the track and field events.

- a) Fixture
- b) Tournament
- c) Athletic Meet
- d) Tour

iii. The physical education officer remains _____ of the athletic meet.

- a) Guest of Honor
- b) President
- c) Secretary
- d) Player

iv. The standard track has _____ lanes.

- a) Both A & B
- b) 08
- c) 07
- d) 06

v. The Marathon race is dedicated to _____ Olympian.

- a) Pheidippides
- b) Zeus
- c) Yosain Bolt
- d) Philops

vi. The staggers are given in the track to _____ the distance for all racers.

- a) Minimize
- b) Equalize
- c) Maximize
- d) Finish

vii. The cross-country race is mostly held on _____.

- a) Standard Track
- b) Roads
- c) Both A & D
- d) Natural Terrain

viii. The long races usually start from _____ km.

- a) 7
- b) 5
- c) 10
- d) 15

ix. The Marathon race can be divided in _____ categories.

- a) 3
- b) 2
- c) 3 & 2
- d) 4

x. The _____ gives the final decision in long distances races.

- a) Chief Judge
- b) Referee
- c) Umpire
- d) Marshal



NATIONAL UNIVERSITY OF PAKISTAN

BS Program (Model Paper)
End-Semester Examination

Course Code: HPE-112

Course Title: Athletics-II (Long Races)

Sign of Invigilator: _____

Time Allowed: 02 Hrs and 40 minutes

Total Marks: 40

General Instructions

- Do not write anything on the question paper other than Registration No.
- Attempt all questions. All the Q&A must be numbered as mentioned in the question paper.

SECTION – B

Q.2. Give short answer to the following:

(5x2=10)

- Define athleticism.
- Differentiate the running and walking.
- Write the short history of Marathon race.
- Why the staggers are given in the standard track.
- Enlist the different phases of a race.

SECTION – C

Write a detailed answer to the following:

(10x3=30)

Q.3. How will you conduct the athletic meet at college level, Explain.

Q.4. What is meant by “obstacle races” also elaborate the rules of the cross country race?

Q.5. Define “start” in running and narrate the duties of officials in long distance races.

**Answer Key**

QUESTION NUMBER	ANSWER
1	D
2	C
3	C
4	B
5	A
6	B
7	D
8	B
9	A
10	B



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN

BS Program (Model Paper)

End-Semester Examination

Course Code: HPE-112L

Time Allowed: 02 Hrs

Course Title: Athletics-II (Long Races) Lab

Total Marks: 30

Sign of Invigilator: _____

General Instructions

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – B**Q.1. Give short answer to the following:****(5x4=20)**

- Draw the standard track with staggers.
- Demonstrate the activities involved up to start in running.
- Sketch the obstacles in a half of the standard track.
- Demonstrate the hurdle crossing used in the obstacle races.

SECTION – C**Answer to the following:****(5x2=10)****Q.2.** Practicable note book.**Q.3.** Viva Voce

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: HPE-113
Course Title: Gymnastic-I
Sign of Invigilator: _____**Time Allowed:** 20 minutes
Total Marks: 10**General Instructions**

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A**Q1. Choose the most appropriate option.****(10*1=10)**

- Who is considered the father of modern gymnastics?**
A) Pierre de Coubertin
B) Johann Friedrich GutsMuths
C) Friedrich Ludwig Jahn
D) Aristotle
- Which ancient civilization influenced early gymnastics training?**
A) Roman
B) Greek
C) Persian
D) Egyptian
- What is the official name of the governing body for gymnastics?**
A) Federation of International Gymnastics
B) International Gymnastics Federation
C) World Gymnastics Association
D) Olympic Gymnastics Committee
- Which of the following is NOT a type of gymnastics?**
A) Artistic gymnastics
B) Rhythmic gymnastics
C) Trampoline gymnastics
D) Marathon gymnastics
- What is the standard diameter of a rhythmic gymnastics hoop?**
A) 50-60 cm
B) 80-90 cm
C) 70-90 cm
D) 60-80 cm
- The width of the balance beam is:**
A) 5 cm
B) 8 cm
C) 10 cm
D) 12 cm
- Which of the following is NOT a skill performed on the pommel horse?**
A) Circles
B) Flairs
C) Handstands
D) Somersaults
- What is the primary purpose of safety mats in gymnastics?**
A) Improve performance
B) Prevent injuries
C) Increase difficulty
D) Reduce competition time
- Which type of mat provides extra bounce for acrobatics?**
A) Foam Mats
B) Spring Mats
C) Crash Mats
D) Folding Mats
- Which grip type is typically used for still rings?**
A) Palm Grips
B) Dowel Grips
C) Rings Grips
D) None of the above



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN**BS Program (Model Paper)****End-Semester Examination****Course Code:** HPE-113**Time Allowed:** 02 Hrs and 40 minutes**Course Title:** Gymnastic-I**Total Marks:** 40**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

SECTION – B**Q.2 Give short answers to the following questions:****(5*2=10)**

- a. What was the significance of gymnastics in ancient Greece?
- b. What are the main disciplines of modern gymnastics?
- c. What material is commonly used for rhythmic gymnastics hoops?
- d. What is the purpose of standardizing equipment dimensions in gymnastics?
- e. Why is safety equipment important in gymnastics training?

SECTION – C**Write detailed answers to the following questions.****(10*3=30)**

- Q.3** Discuss the historical evolution of gymnastics from ancient civilizations to modern times.
- Q.4** Explain in detail the different types of gymnastics and their unique characteristics.
- Q. 5** Explain the different types of gymnastics mats used for safety and training, including their dimensions and purposes.

**Answer Key**

QUESTION NUMBER	ANSWER
1	C
2	B
3	B
4	D
5	C
6	C
7	D
8	B
9	B
10	C



NATIONAL UNIVERSITY OF PAKISTAN

BS Program (Model Paper)

End-Semester Examination

Reg.No: _____

Course Code: ISL-102

Time Allowed: 20 minutes

Course Title: Arabic Language-I

Total Marks: 10

Sign of Invigilator: _____

ہدایات: حذف کر کے / کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ براہ کرم لیڈ پینسل کا استعمال نہ کریں۔
حصہ اول (حصہ معروضی) کے تمام اجزاء کا جواب اسی پرچے پر دیں۔

حصہ اول

1. عربی زبان کی ابتدا کب ہوئی؟
2. ماضی اور مضارع کیا ہیں؟ الف) حروف کی اقسام (ب) فعل کی اقسام (ج) اسم کی اقسام (د) نحو کے اصول
3. فعل ناقص "کس کو کہتے ہیں؟
4. الف) وہ فعل جو مکمل معنی دے (ب) وہ فعل جو فعل امر میں آتا ہے (ج) وہ فعل جو ہمیشہ مضارع میں آتا ہے (د) وہ فعل جو اسم کے بغیر مکمل نہ ہو اسم فاعل کی پہچان کیا ہے؟
5. الف) وہ اسم جو فعل پر عمل کو ظاہر کرے (ب) وہ اسم جو فعل کے کرنے والے کو ظاہر کرے (ج) وہ اسم جو زمانہ اور مکان کو ظاہر کرے (د) وہ اسم جو حرف نداء کے ساتھ آتا ہو حروف نداء کا استعمال کس کے لیے ہوتا ہے؟
6. الف) کسی کو نصیحت کرنے کے لیے (ب) کسی کو روکنے کے لیے (ج) کسی کو بلانے کے لیے (د) کسی چیز کی وضاحت کرنے کے لیے اجوف کو لغت میں کیا کہتے ہیں؟ الف) دم کٹے کو (ب) خالی پیٹ کو (ج) لپیٹے ہوئے کو (د) دو چند کو
7. عربی گرائمر میں مضارع کس معانی کے لیے آتا ہے؟ الف) ماضی کے لیے (ب) مستقبل کے لیے (ج) حال کے لیے (د) حال و مستقبل دونوں کے لیے
8. حروف جوازم کتنے ہیں؟ الف) 3 (ب) 5 (ج) 7 (د) 9
9. حروف نواصب کتنی ہیں؟ الف) 4 (ب) 3 (ج) 5 (د) 2
10. ثلاثی مجرد کے کل باب کتنے باب ہیں؟ الف) 3 (ب) 5 (ج) 6 (د) 7

**NATIONAL UNIVERSITY OF PAKISTAN**

BS Program (Model Paper)

End-Semester Examination

Reg.No: _____

Course Code: ISL-102

Time Allowed: 02 Hrs and 40 minutes

Course Title: Arabic Language-I

Total Marks: 40

Sign of Invigilator: _____

ہدایات: حذف کر کے / کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ براہ کرم لیڈ پیسل کا استعمال نہ کریں۔
حصہ اول (حصہ معروضی) کے تمام اجزاء کا جواب اسی پرچے پر دیں۔

حصہ دوم: مختصر جوابات

1. عربی زبان کو اسلامی علوم میں کیوں اہمیت حاصل ہے؟

((5x2=10))

2. حروف نداء کے تین الفاظ لکھیں اور ان کا استعمال بتائیں؟

3. حروف جوازم کی تعریف کریں اور ان کی تعداد بمع امثال لکھیں؟

4. صحیح اور مہموز کی تعریف بمع امثال لکھیں؟

5. مثال، اجوف اور ناقص کی تعریف کریں؟

حصہ سوم: تفصیلی سوالات (30 نمبر)**(10x3=30)**

1. عربی زبان کے ارتقاء اور اس کی اسلامی علوم میں اہمیت پر تفصیل سے روشنی ڈالیں؟

2. ثلاثی مزید فیہ کی تعریف کریں نیز ان کے ابواب کے نام لکھیں اور کم از کم دو مثالیں لکھیں؟

3. نصر سے فعل ماضی معلوم کی گردان لکھیں؟

**Answer key**

QUESTION NUMBER	ANSWER
1	ج
2	ب
3	د
4	الف
5	ج
6	ب
7	د
8	ب
9	الف
10	ج



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN
BS Program (Model Paper)
End-Semester Examination

Course Code: ISL-103

Time Allowed: 20 minutes

Course Title: History of Fiqh

Total Marks: 10

Sign of Invigilator: _____

General Instructions:

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A

Q1. Choose the most appropriate option.

(10*1=10)

- 1 What is the literal meaning of *Fiqh*?
 - f) Wisdom
 - b) Deep understanding
 - c) Talk
 - d) Hadith
- 2 What was the basis of legislation during the Prophetic era?
 - a) Hanafi jurisprudence
 - b) Royal decrees
 - c) Customs and traditions
 - d) Quran and Sunnah
- 3 "Al-Ziyādāt" belongs to which school of jurisprudence?
 - a) Ḥanafī
 - b) Shāfi'ī
 - c) Mālikī
 - d) Ḥanbalī
- 4 The source of Roman law was:
 - a) The Quran
 - b) Hadith
 - c) Ḥanafī jurisprudence
 - d) Royal decrees
- 5 The famous book of Imam Mālik is:
 - a) Al-Muwaṭṭa'
 - b) Al-Risālah
 - c) Al-Umm
 - d) None
- 6 The founder of the Ḥanafī school of thought was:
 - a) Imam Abū Ḥanīfah
 - b) Imam Ja'far
 - c) Imam Muḥammad ibn Idrīs
 - d) Imam Aḥmad
- 7 When was the Islamic Ideology Council established?
 - a) 1956
 - b) 1962
 - c) 1975
 - d) 1985
- 8 Which era is considered the golden period in the history of Islamic law?
 - a) Era of Abū Bakr (Ṣiddīq)
 - b) Umayyad era
 - c) Abbasid era
 - d) Present era
- 9 What was the purpose of establishing the Federal Shariah Court?
 - a) Modern legislation
 - b) Prevention of un-Islamic laws
 - c) Implementation of un-Islamic laws
 - d) None
- 10 In which constitution was the *Objectives Resolution* made a part of the preamble?
 - a) 1956
 - b) 1962
 - c) 1973
 - d) All of these



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN
BS Program (Model Paper)
End-Semester Examination

Course Code: ISL-103

Course Title: History of Fiqh

Sign of Invigilator: _____

Time Allowed: 02 Hrs and 40 minutes

Total Marks: 40

General Instructions

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the Q&As must be numbered as mentioned in the question paper.

SECTION – B

Q.2 Give short answers to the following questions:

(5*2=10)

- 1 Explain *Shariah* and *Usul al-Fiqh*.
- 2 What is the difference between Islamic law and Roman law?
- 3 Write two jurisprudential principles of Ḥanbalī school of *Fiqh*.
- 4 What is the difference between the jurisprudential principles of Imam Mālik and Imam al-Shāfiʿī?
- 5 Explain the role of the Islamic Ideology Council.

SECTION – C

Write a detailed answer to the following questions.

(10*3=30)

- Q.3** Explain the meaning of *Fiqh*. Also highlight its characteristics.
- Q.4** Write an essay on “Introduction to the Ḥanafī School of Jurisprudence.”
- Q.5** Elaborate on the Islamic provisions of the Constitution of Pakistan 1973.



NATIONAL UNIVERSITY OF PAKISTAN
BS Program (Model Paper)
End-Semester Examination

Reg.No: _____

Course Code: ISL-103
Course Title: History of Fiqh
Sign of Invigilator: _____

Time Allowed: 20 minutes
Total Marks: 10

ہدایات:

حذف کر کے/کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ براہ کرم لیڈ پینسل کا استعمال نہ کریں۔ حصہ اول کے تمام اجزاء کا جواب اسی پرچے پر دیں۔

حصہ اول

(10x1=10)

سوال نمبر 1: مناسب ترین جواب کا انتخاب کریں۔

1. فقہ کا لغوی معنی کیا ہے؟
 (ا) حکمت (ب) گہری سمجھ (ج) بات چیت (د) حدیث
2. عہد نبوی میں قانون سازی کی کیا بنیاد تھی؟
 (ا) فقہ حنفی (ب) فرامین شاہی (ج) رسم و رواج (د) قرآن و سنت
3. "الزیادات" کس فقہی مکتب کی کتابیں ہیں؟
 (ا) حنفی (ب) شافعی (ج) مالکی (د) حنبلی
4. رومن قانون کا ماخذ تھا۔
 (ا) قرآن (ب) حدیث (ج) فقہ حنفی (د) فرامین شاہی
5. امام مالک کی مشہور کتاب ہے:
 (ا) المؤطا (ب) الرسالة (ج) الام (د) کوئی نہیں
6. فقہ حنفی کے بانی تھے:
 (ا) امام ابو حنیفہ (ب) امام جعفر (ج) امام محمد بن ادریس (د) امام احمد
7. اسلامی نظریاتی کونسل کب قائم ہوئی؟
 (ا) 1956ء (ب) 1962ء (ج) 1975ء (د) 1985ء
8. تاریخ فقہ اسلامی کا سب سے سنہری دور کون سا ہے؟
 (الف) عہد صدیقی (ب) عہد بنو امیہ (ج) عہد بنو عباس (د) موجودہ دور
9. وفاقی شرعی عدالت کی تشکیل کا مقصد کیا ہے:
 (الف) جدید قانون سازی (ب) غیر شرعی قوانین کی روک تھام (ج) غیر شرعی قوانین کا نفاذ (د) کچھ نہیں
10. قرارداد مقاصد کو کس آئین میں دیباچے کا حصہ بنایا گیا؟
 (ا) 1956ء (ب) 1962ء (ج) 1973ء (د) سب کا



NATIONAL UNIVERSITY OF PAKISTAN
BS Program (Model Paper)
End-Semester Examination

Reg.No: _____

Course Code: ISL-103

Time Allowed: 2 Hrs and 40 minutes

Course Title: History of Fiqh

Total Marks: 40

Sign of Invigilator: _____

ہدایات:

سوالیہ پرچے پر رول نمبر کے علاوہ کچھ نہ لکھیں۔ تمام سوالات اور جوابات کو سوالیہ پرچے کے مطابق نمبر دیں۔

حصہ دوم

سوال نمبر 2۔ مندرجہ ذیل سوالات کے مختصر جوابات تحریر کریں۔
 (5X2=10)

- i. شریعہ اور اصول فقہ کی وضاحت کریں۔
- ii. اسلامی قانون اور رومن قانون میں کیا فرق ہے؟
- iii. حنبلی مکتب فقہ کے دو فقہی اصول تحریر کیجئے۔
- iv. امام مالک اور امام شافعی کے فقہی اصولوں میں کیا فرق ہے؟
- v. اسلامی نظریاتی کونسل کا کردار واضح کریں۔

حصہ سوم

سوال نمبر 3۔ فقہ کا مفہوم واضح کرتے ہوئے اس کی خصوصیات پر روشنی ڈالیں۔
 (10)

سوال نمبر 4۔ "حنفی مکتب فقہ کا تعارف" پر مضمون تحریر کریں۔
 (10)

سوال نمبر 5۔ دستور پاکستان 1973ء کی اسلامی دفعات واضح کریں۔
 (10)



Answer Keys

QUESTION NUMBER	ANSWERS
1	B
2	D
3	A
4	D
5	A
6	A
7	B
8	C
9	B
10	C

QUESTION NUMBER	ANSWERS
1	(ب)
2	(د)
3	(ا)
4	(د)
5	(ا)
6	(ا)
7	(ب)
8	(ج)
9	(ب)
10	(ج)

**NATIONAL UNIVERSITY OF PAKISTAN**

BS Program (Model Paper)
End-Semester Examination

Reg.No: _____

Course Code: MATH-112

Time Allowed: 20 minutes

Course Title: Elementary Set Theory

Total Marks: 10

Sign of Invigilator: _____

General Instructions

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A**Q.1 Encircle the most appropriate answer.****(10*1=10)**

- A set S is said to be ----- if it has the same cardinality as a proper subset of itself.
 - finite
 - infinite
 - countable
 - uncountable
- Let a and b are elements of Partially ordered set S . We say a and b are-----
 If $a \leq b$ or $b \geq a$ (i.e If one of them precedes other)
 - non comparable
 - comparable
 - minimal
 - maximal
- Every finite Partially ordered set has at least one ----- element
 - first element
 - minimal element
 - last element
 - supremum
- Set of Irrational numbers $\mathbb{Q}$ is -----
 - denumerable
 - countable
 - non denumerable
 - uncountable
- An infinite sequence with distinct elements is
 - finite
 - infinite
 - denumerable
 - none of these
- Let $z = \{\dots - 3, - 2, - 1, 0, 1, 2, 3, \dots\}$ Then initial segment $s(0) = --$
 - $\{1, 2, 3, \dots, 10\}$
 - $\{\dots, - 3, - 2, - 1\}$
 - $\{\dots, - 3, - 2, - 1, 0\}$
 - none of these
- Let S be a Partially ordered set. An element $a \in S$ is called maximal element of S if
 - $x \leq a \forall x \in S$
 - $x \geq a \forall x \in S$
 - $a \leq x$ for $x \in S$ implies $x = a$
 - $a \geq x$ for $x \in S$ implies $x = a$
- If $X_1 \subseteq Y \subseteq X$ or $X \sim X_1$ then
 - $X_1 \sim Y$
 - $X \sim Y$
 - $X \leq Y$
 - $X_1 \sim X$
- $U \in A$ is said to be upper bound of B if
 - $x \geq U \forall x \in B$
 - $x \leq U \forall x \in B$
 - $U \leq x \forall x \in A$
 - None of these
- Let $A = \{x \in \mathbb{R} : 0 < x < 1\}$ and $a, b \in A, a \leq b$ if $a \leq b$ then first element of A is
 - 0
 - 1
 - 1
 - do not exist

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)****End-Semester Examination**

Reg.No: _____

Course Code: MATH-112**Time Allowed:** 02 Hrs and 40 minutes**Course Title:** Elementary Set Theory**Total Marks:** 40**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

SECTION – B**Q.2 Give short answer to the following questions.****(5*2 = 10)**

- Prove that $[0, 1]$ is non denumerable.
- Prove that for any set A , $A \leq 2^A$.
- Let $A = \{1, 2, 3, 4, 6, 8, 9, 12, 18, 24\}$ be ordered by relation " x divides y ". Draw the Hasse Diagram of A . Find its maximal and minimal elements.
- Show by example that commutative laws under addition do not hold for ordinal addition.
- Define Zorn's lemma and show that every vector V space admits a basis.

SECTION – C**Write answers to the following questions.****(10*3=30)**

- Q.3 a)** For any cardinals α, β, γ show that $\alpha^\beta \alpha^\gamma = \alpha^{\beta+\gamma}$ (5)
- b)** Consider set Q of rational numbers with the usual order $\leq$ and consider subset D of Q denoted by $D = \{x \in Q: 8 < x^3 < 15\}$. Is D is bounded above or below? Do $\sup(D)$ and $\inf(D)$ exist? (5)
- Q.4 a)** Prove that the relation of Similarity is an equivalence relation. (5)
- b)** Let $s(A)$ denote the collection of all initial segments of elements in a well-ordered set A and let $S(A)$ be ordered by set inclusion. Then A is similar to $S(A)$. (5)
- Q.5 a)** Show that axiom of choice is equivalent to Zermelo's postulate. (5)
- b)** Describe Cantor's paradox and Russell's paradox. (5)

**Answer Key**

QUESTION NUMBER	ANSWERS
1	B
2	B
3	B
4	C
5	C
6	B
7	C
8	B
9	B
10	D

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: MGT-131**Time Allowed:** 20 minutes**Course Title:** Financial Accounting I**Total Marks:** 10**Sign of Invigilator:** _____**General Instructions**

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A**Q1. Choose the most appropriate option.(10x1=10)**

- Which of the following is NOT a major area of accounting?
 - Financial Accounting
 - Cost Accounting
 - Medical Accounting
 - Tax Accounting
- GAAP refers to:
 - Government Approved Accounting Practices
 - Generally Accepted Accounting Principles
 - General Assessment of Accounting Procedures
 - Global Accounting and Auditing Principles
- The double-entry bookkeeping system is based on:
 - Only assets and liabilities
 - The accounting equation
 - Recording expenses only
 - Recognizing revenue when cash is received
- A trial balance is prepared to:
 - Record transactions
 - Identify errors in accounts
 - Show financial position
 - Prepare financial statements
- The primary purpose of an income statement is to:
 - Show financial position
 - Show revenue and expenses over a period
 - Record all cash transactions
 - Show only company assets
- The statement of owner's equity includes:
 - Only net income
 - Only assets and liabilities
 - Owner's capital and withdrawals
 - Operating expenses only
- Adjusting entries are made to:



-
- a. Close temporary accounts
 - b. Recognize expenses and revenues in the correct period
 - c. Record all cash transactions
 - d. Prepare journal entries
- viii.** Bank reconciliation is used to:
- a. Detect errors and fraud in cash transactions
 - b. Prepare an income statement
 - c. Adjust capital accounts
 - d. Close the ledger accounts
- ix.** Which of the following is considered capital expenditure?
- a. Payment of salaries
 - b. Purchase of machinery
 - c. Rent expense
 - d. Office supplies expense
- x.** The Global Reporting Initiative (GRI) framework is used for:
- a. Preparing tax reports
 - b. Sustainability reporting
 - c. Profit calculation
 - d. Bank reconciliation

**NATIONAL UNIVERSITY OF PAKISTAN**

Reg.No: _____

BS Program (Model Paper)**End-Semester Examination****Course Code:** MGT-131**Time Allowed:** 02 Hrs and 40 minutes**Course Title:** Financial Accounting I**Total Marks:** 40**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Registration No.
- Attempt all questions. All the Q&A must be numbered as mentioned in the question paper.

SECTION – B**Q.2 Give short answer to the following:****(5x2=10)**

- Define accounting and explain its purpose in business.
- Explain the importance of adjusting entries in financial reporting.
- What is a subsidiary ledger, and how does it help in controlling accounts?
- Differentiate between capital and revenue expenditures with examples.
- What is sustainability accounting, and why is it important in today's financial world?

SECTION – C**Write detailed answers of the following questions.****(3x10=30)****Q.3. a)** ABC Enterprises completed the following transactions in April 2025: **(2+2+1=5)**

- Apr 1: Started business with \$80,000 in cash.
- Apr 4: Purchased inventory for \$25,000 on credit.
- Apr 8: Sold goods for \$18,000 (cost \$10,500) for cash.
- Apr 15: Paid office rent of \$3,500.

Required:

- Record the **journal entries** for these transactions.
- Prepare a **trial balance** for the business.

b) XYZ Company provided the following year-end information: **(2+2+1=5)**

- Office Supplies Expense (adjusted) = \$2,000
- Depreciation Expense on Equipment = \$5,000
- Unearned Revenue (adjusted) = \$3,500
- Accrued Salaries Expense = \$4,000

Required:

- Pass the necessary **adjusting journal entries**.
- Prepare the **Income Statement** and **Statement of Owner's Equity**

Q.4. a) The cash book and bank statement of ABC Traders show the following discrepancies: **(3+2=5)**

- Balance as per cash book: \$30,000



- ii. Outstanding checks: \$6,000
- iii. Deposits in transit: \$4,500
- iv. Bank charges deducted by bank: \$800

Required:

Prepare a **bank reconciliation statement**.

- b) A business incurs the following expenses during the year:
- i. Purchase of a delivery van for \$50,000
 - ii. Repairs and maintenance of machinery for \$3,500
 - iii. Purchase of office furniture for \$10,000
 - iv. Payment of rent for \$6,000

Required:

Classify each expense as **capital or revenue expenditure** with justification.

Q.5

Explain the role of sustainability accounting and Global Reporting Initiative (GRI) in modern financial reporting.

(5+5=10)

**Answer Key**

QUESTION NUMBER	ANSWER
1	C
2	B
3	B
4	B
5	B
6	C
7	B
8	A
9	B
10	B



Reg.No: _____

National University of Pakistan**BS Program (Model Paper)****End-Semester Examination****Course Title: Physics Lab-II (Wave and Optics)****Time Allowed: 03 Hrs****Course Code: PHY-113 (L)****Total Marks: 50****Sign of Invigilator: _____****General Instructions**

- Do not write anything on the question paper other than Registration No.
- Attempt all questions. All the Q&A must be numbered as mentioned in the question paper.

NOTE: - Attempt any one question from 1st three question as allotted by examiner. All questions carry equal marks. Solve all of following parts for allotted questions.
Question No. 4 is compulsory.

- a- Write required apparatus. (2)
- b- Write precise procedure and theory. (10+5)
- c- Observation and calculations. (15)
- d- Diagram / Graph. (5)
- e- Experimental results. (3)

Q. No 1 : - Determine the frequency of AC supply by Melde's method.

Q. No 2: - Determine the velocity of sound by Kundt's tube.

Q. No 3 : - Determine the height of given object by sextant.

Q. No 4:- Viva Voice and practical manual. (5+5)



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN**BS Program (Model Paper)
End-Semester Examination****Course Code: POL-103****Time Allowed: 20minutes****Course Title: Introduction to Political Science II****Total Marks: 10****Sign of Invigilator:****General Instructions**

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION –A**Q1. Choose the most appropriate option. (10x1=10)****1. Which of the following is a key characteristic of a unitary government?**

- a) Power is shared between central and state governments
- b) All powers are concentrated in the central government
- c) States have more power than the central government
- d) It has two houses of the legislature

2. Which of the following is an example of a federation?

- a) United Kingdom
- b) France
- c) United States
- d) China

3. Which of the following is a fundamental principle of democracy?

- a) Rule by a single leader
- b) Suppression of opposition
- c) Free and fair elections
- d) Centralized decision-making

4. Authoritarian governments are characterized by:

- a) Free and fair elections
- b) Limited political freedoms
- c) Decentralization of power
- d) Strong opposition parties

5. In a parliamentary system, the head of government is usually:

- a) Directly elected by the people
- b) A monarch
- c) The leader of the majority party in parliament
- d) Appointed by the judiciary



6. **Which of the following is a feature of a presidential government?**
 - a) The executive is dependent on the legislature
 - b) The president serves as both head of state and head of government
 - c) The prime minister has more power than the president
 - d) The government is accountable to the parliament
7. **What is a constitution?**
 - a) A document that outlines the fundamental laws and principles of a state
 - b) A temporary set of rules for political parties
 - c) A law regulating businesses and trade
 - d) A system of local government
8. **Which of the following is NOT a recognized source of law?**
 - a) Religious principles
 - b) Personal opinions of judges
 - c) Legislation
 - d) Customary practices
9. **Political parties play an important role in democracy by:**
 - a) Suppressing opposition voices
 - b) Selecting candidates for elections and forming governments
 - c) Preventing political debates
 - d) Limiting citizen participation in governance
10. **Public opinion is primarily formed through:**
 - a) Government-imposed regulations
 - b) Media, education, and social interactions
 - c) Legal restrictions on speech
 - d) Military enforcement of views

**Answer Key**

QUESTION NUMBER	ANSWER
1	B
2	C
3	C
4	B
5	C
6	B
7	A
8	B
9	B
10	B



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN

BS Program (Model Paper)

End-Semester Examination

Course Code: POL-103

Time Allowed: 2 Hours and 40 minutes

Course Title: Introduction to Political Science II

Total Marks: 40

Sign of Invigilator: _____

General Instructions

- Do not write anything on the question paper other than Registration No.
- Attempt all questions. All the Q&A must be numbered as mentioned in the question paper.

SECTION-B

Q.2. Give short answer to the following:

(5x2=10)

- What are the key features of a presidential system? (Write any three)
- Define a constitution.
- Name two sources of law.
- Define parliamentary system.
- How is public opinion shaped in a society?

SECTION-C

Write a detailed answer to the following

(10x3=30)

Q.3. Define democracy. What are the key principles of a democratic system, and why is it considered the best form of government?

(2+5+3)

Q.4. Describe an authoritarian government. How does it impact individual freedoms and political participation? Give historical examples

(5+5)

Q.5 What are political parties? Discuss their functions and significance in a democratic system.

(5+5)



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN**BS Program (Model Paper)
End-Semester Examination****Course Code:** POL-111**Time Allowed:** 20 minutes**Course Title:** Constitutional Development in Pakistan**Total Marks:** 10**Sign of Invigilator:** _____**General Instructions**

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION –A**Q1. Choose the most appropriate option.****(10x1=10)**

- _____ had abrogated the 1956 Constitution of Pakistan.
 - Ayub Khan
 - Sikandar Mirza
 - Malik Ghulam Muhammad
 - None of these
- According 1962 Constitution President was elected by _____.
 - Senate
 - Parliament
 - National Assembly
 - Basic Democrats
- Parliament of Pakistan has power to _____ the budget.
 - present
 - prepare
 - pass
 - none of these
- 1973 Constitution of Pakistan was implemented on _____.
 - 23rd March 1973
 - 14th Aug 1973
 - 12th March 1973
 - 16th October 1973
- Chief Justice of Pakistan take oath before _____.
 - Prime Minister
 - President
 - Speaker of National Assembly
 - Chairman of Senate
- _____ act as representative of federal government in province.
 - Governor
 - President
 - leader of opposition
 - Chief Minister
- 13th amendment had retrieved the power of _____.
 - President
 - Prime Minister
 - Governor
 - Chief Justice of Pakistan
- NWFP was renamed as KPK through the _____ amendment.
 - 8th
 - 13th
 - 17th
 - 18th
- By _____ majority Parliament of Pakistan can amend the constitution
 - 1/3
 - 1/2
 - 2/3
 - 3/4
- According 1956 Constitution Pakistan had _____ federating units.
 - Two
 - Four
 - Five
 - Four and FATA

**Answer Key**

QUESTION NUMBER	ANSWER
1	B
2	D
3	C
4	B
5	B
6	A
7	B
8	D
9	C
10	A



NATIONAL UNIVERSITY OF PAKISTAN

BS Program (Model Paper)

End-Semester Examination

Reg.No: _____

Course Code: POL-111

Time Allowed: 2 Hrs and 40 minutes

Course Title: Constitutional Development in Pakistan

Total Marks: 40

Sign of Invigilator: _____

General Instructions

- Do not write anything on the question paper other than Registration No.
- Attempt all questions. All the Q&A must be numbered as mentioned in the question paper.

SECTION-B

Q.1. Give short answer to the following:

(5x2=10)

- a) How can the Prime Minister of Pakistan be removed from the office?
- b) Write importance of Objective Resolution?
- c) Discuss central legislature according to the 1962 Constitution?
- d) What does FATA stand for? When was it annexed to KPK?
- e) What 8th and 17th Amendments are known for?

SECTION-C

Write a detailed answer to the following.

(10x3=30)

Q.3. Explain Basic Principles Committee (BPC) and its Three Reports?

(2+8)

Q.4. What were the roles of Malik Ghulam Muhammad and Sikandar Mirza in constitutional history of Pakistan?

(5+5)

Q.5. How is the President elected? What are the functions of the President according to the 1973 Constitution of Pakistan?

(5+5)



NATIONAL UNIVERSITY OF PAKISTAN

**BS Program (Model Paper)
End-Semester Examination**

Reg.No: _____

Course Code: PSY-103

Course Title: History and Schools of Psychology

Sign of Invigilator: _____

Time Allowed: 20 minutes

Total Marks: 10

General Instructions

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A

Q1. Choose the most appropriate option. (10*1=10)

- Which of the following is considered a persistent question in psychology?
 - Nature vs. nurture
 - Left-handed vs. right-handed dominance
 - The best method of memorization
 - The role of astrology in behaviour
- Who is known as the "Father of Medicine" in early Greek history?
 - Plato
 - Socrates
 - Hippocrates
 - Aristotle
- Muslim scientists emphasized which aspect in their study of psychology?
 - Materialism
 - Spirit and soul
 - Only behaviour
 - Structuralism
- Which Greek philosopher was known for his method of questioning to gain knowledge?
 - Aristotle
 - Plato
 - Socrates
 - Epicurus
- Plato's philosophy mainly focused on:
 - Empirical observation
 - Materialism
 - The theory of forms
 - Behaviorism
- Skepticism as a philosophical school emphasizes:
 - Absolute truth
 - Doubt and questioning
 - Faith and spirituality
 - Emotional reasoning



Vii. Who is credited with founding the first psychology laboratory?

- a) B.F. Skinner
- b) Wilhelm Wundt
- c) Sigmund Freud
- d) John Watson

viii. B.F. Skinner is best known for his work in:

- a) Structuralism
- b) Functionalism
- c) Behaviourism
- d) Humanistic psychology

ix. Structuralism was primarily concerned with:

- a) Understanding the function of the mind
- b) The structure of consciousness
- c) The role of emotions in behaviour
- d) The application of psychology in real life

x. Which school of thought focused on how mental processes help individuals adapt to their environment?

- a) Structuralism
- b) Behaviourism
- c) Functionalism
- d) Gestalt psychology

**Answer Key**

QUESTION NUMBER	ANSWER
1	A
2	C
3	B
4	C
5	C
6	B
7	B
8	C
9	B
10	C

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)****End-Semester Examination**

Reg.No: _____

Course Code: PSY-103**Time Allowed:** 02 Hrs and 40 minutes**Course Title:** History and Schools of Psychology**Total Marks:** 40**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

SECTION – B**Q.2 Give short answer to the following:****(5*2=10)**

- Compare and contrast the views of Socrates and Aristotle on human behaviour.
- Briefly describe Wilhelm Wundt's contribution to psychology.
- Differentiate between pure, applied, and scientific psychology.
- Briefly explain the psychological aspects of artificial intelligence with example from everyday life
- What factors contribute to psychology's recognition as a science? Briefly explain

SECTION – C

Write a detailed answer of the following questions

(10*3=30)**Q.3** Describe the early development of psychology. Discuss how Greek philosophers, Muslim scientists, and early psychologists contributed to shaping psychology.**Q.4** Compare and contrast structuralism, functionalism, and behaviorism. Explain their key principles and major contributors in psychology.**Q 5.** Analyze the recent trends in psychology worldwide and in Pakistan. Discuss how psychology is evolving as a scientific discipline and its significance in today's world.



NATIONAL UNIVERSITY OF PAKISTAN
BS Program (Model Paper)
End-Semester Examination

Reg.No: _____

کورس کوڈ: URDU-112

منٹ: 20 کل وقت

کورس ٹائٹل: اردو زبان قواعد و املا

کل نمبر: 10

دستخط نگران

ضروری ہدایات: کاٹ کر دوبارہ اس پرچے کو صرف اسی سوالیہ کاپی پر حل کر کے نگران کے حوالے کریں۔

لکھنے کی اجازت نہیں لیڈ پینسل کا استعمال ممنوع ہے۔

حصہ اول

(10 X 1=10)

درست جواب کا انتخاب کیجیے۔

- 1۔ "اردو املا" کس کی تصنیف ہے؟
 ا. شبیر حسن خان
 ب. رشید حسن خان
 ج. رشید امجد
 د. آن میں سے
- 2۔ "الشمس" کا شمار
 ا. حروف شمسی
 ب. حروف قمری
 ج. ڈونوں میں
 د. آن میں سے
- 3۔ علم صرف میں بحث ہوتی ہے؟
 ا. اقدار سے
 ب. الفاظ سے
 ج. افکار سے
 د. اقوام سے
- 4۔ دو یا دو سے زیادہ بامعنی الفاظ کا مجموعہ جس سے کہنے والے کا مقصد پورا ہو جائے کہلاتا ہے؟
 ا. مرکب اضافی
 ب. مرکب عطفی
 ج. مرکب ناقص
 د. مرکب تام
- 5۔ "قواعد اردو" کس کی تصنیف ہے؟
 ا. شبیر حسن خان
 ب. رشید حسن خان
 ج. مولوی عبدالحق
 د. محمد حسین آزاد
- 6۔ دو یا دو سے زیادہ الفاظ کا ایسا مجموعہ جس سے حقیقی بجائے مجازی معنی مراد لیے جائیں کہلاتا ہے؟
 ا. روزمرہ
 ب. محاورہ
 ج. مصدر
 د. اسم
- 7۔ غلط العام سے مراد وہ لفظ یا ترکیب ہے جو غلط ہونے کے باوجود ...؟
 ا. مستعمل ہو
 ب. مستعمل نہ ہو
 ج. احتیاط سے استعمال کیا جائے
 د. کثرت سے استعمال کیا جائے
- 8۔ لکھنا، پڑھنا، ہنسنا، بولنا قواعد کی رو سے کیا ہیں؟
 ا. مصدر
 ب. اسم
 ج. فاعل
 د. مفعول
- 9۔ "علم بجا" علم ہے؟
 ا. حروف کی آواز اور حرکات و سکنات کا
 ب. حروف کی ترسیمی صورت کا
 ج. حروف کے استعمال کا
 د. حروف کے مفہیم کا
- 10۔ درج ذیل لفظ کے درست املا کی نشاندہی کریں؟
 ا. لاپروا
 ب. لاپرواہ
 ج. لاپرواح
 د. لاپرواں



NATIONAL UNIVERSITY OF PAKISTAN
BS Program (Model Paper)
End-Semester Examination

Reg.No: _____

منٹ: کل وقت 2

کل نمبر: 40
دستخط نگران

کورس کوڈ: URDU-112

گھنٹے 40 منٹ
کورس ٹائٹل: اردو زبان قواعد و املا**(حصہ دوم (مختصر جوابات**سوال نمبر 2: مندرجہ ذیل سوالوں کے مختصر جوابات تحریر کریں۔
(5x2=10)

- 1۔ اسم معرفہ اور اسم نکرہ کی تعریف بیان کریں۔
 - 2۔ علم نحو کی وضاحت کریں۔
 - 3۔ حروف ابجد کسے کہتے ہیں واضح کریں۔
 - 4۔ ہرروف شمسی و حروف قمری میں فرق بیان کریں۔
 - 5۔ درج ذیل الفاظ کا درست املا تحریر کریں۔
 - 1۔ گزارش 2۔ درستگی 3۔ چوہدری 4۔ برائے مہربانی 5۔ کاروائی 6۔ بلکل 7۔ اسلام و علیکم 8۔ حلوہ و قوس و قزح 10۔ طلباء
- حصہ سوم

(10x3=30)

سوال نمبر 3 (ا): مرکب کسے کہتے ہیں؟ مرکبات کی اقسام بیان کریں۔

(05)

(ب): روزمرہ اور محاورہ میں فرق بیان کرتے ہوئے دونوں کی پانچ پانچ مثالیں تحریر کریں۔

(05)

سوال نمبر 4: اردو قواعد سے متعلق ابتدائی کام کا جائزہ پیش کریں۔

(10)

سوال نمبر 5: اردو املا کے مسائل اور ان کی اہمیت بیان کریں۔



حصہ اول درست جوابات

ا	1
ا	2
ب	3
د	4
ج	5
ب	6
ا	7
ا	8
ا	9
ا	10



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN
BS Program (Model Paper)
End-Semester Examination

Course Code: URDU-122

Time Allowed: 20 minutes

Course Title: بیان و بدیع و عروض

Total Marks: 10

Sign of Invigilator: _____

حصہ اول (حصہ حذف کر کے / کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پینسل کا استعمال نہ کریں۔
معروضی) کے تمام اجزاء کا جواب اسی پرچے پر دیں۔

حصہ اول

(10*1=10)

سوال نمبر 1: مناسب ترین جواب کا انتخاب کریں۔

۱. علم بیان کا موضوع کیا ہے؟
(ا) لفظ (ب) معنی (ج) زبان (د) بلاغت
۲. علم بیان کا دارومدار کتنی چیزوں پر ہے؟
(ا) پانچ (ب) چھ (ج) دو (د) چار
۳. "جانے نہ جانے گل ہی نہ جانے باغ تو سارا جانے ہے" میں "گل" کیا ہے؟
(ا) مشبہ (ب) استعارہ (ج) مجاز مُرسل (د) تشبیہ
۴. "حسن تعلیل" سے مراد ہے؟
(ا) علیل ہونا (ب) علت بیان کرنا (ج) علالت بیان کرنا (د) عمل کرنا
۵. تقطیع میں کون سا طریقہ اختیار کیا جاتا ہے؟
(ا) مکتوبی (ب) ملفوظی (ج) لغوی (د) اصلاحی
۶. مثنیٰ کے معنی ہیں؟
(ا) سامنے والا (ب) بعد میں آنے والا (ج) ہشت پہلو (د) چہار پہلو
۷. تقطیع میں الف ممدوہ (آ) جب کسی کلمے سے شروع ہو تو شمار ہوتا ہے۔
(ا) دو حرف کے (ب) ایک کے برابر (ج) تین کے برابر (د) شمار نہیں ہوتا
۸. جب پہلا اور تیسرا حرف متحرک اور دوسرا ساکن ہو تو ----- کہلاتا ہے۔
(ا) سبب ثقیل (ب) وتد مفروق (ج) فاصلہ (د) سبب خفیف
۹. " نف " کا کیا مطلب ہے؟



(د) ان میں سے کوئی بھی نہیں

(ج) بیان کرنا

(ب) لپیٹنا

(ا) نشر کرنا

۱۰۔ "بحر الفصاحت" کے مصنف کون ہیں؟

(د) عابد علی عابد

(ج) نجم الغنی

(ب) حبیب اللہ

(ا) آغا صادق



Reg.No: _____

NATIONAL UNIVERSITY OF PAKISTAN
BS Program (Model Paper)
End-Semester Examination

Course Code: URDU-122

Time Allowed: 2 Hrs and 40 minutes

Course Title: بیان و بدیع و عروض

Total Marks: 40

Sign of Invigilator: _____

ہدایات

سوالیہ پرچے پر رول نمبر کے علاوہ کچھ نہ لکھیں۔ تمام سوالات اور جوابات کو سوالیہ پرچے کے مطابق نمبر دیں۔

حصہ دوم

(5x2 = 10)

نمبر 2 : مندرجہ ذیل سوالات کے مختصر جوابات تحریر کریں۔

- i. نعت مراعاة النظیر سے کیا مراد ہے؟ شعری مثال تحریر کریں۔
- ii. تقطیع کے کوئی سے دو اصول تحریر کریں۔
- iii. بحر رمل کی تعریف کریں۔
- iv. تجنیس تام کی تعریف کریں؟
- v. "سب کہاں کچھ لالہ و گل میں نمایاں ہو گئیں خاک میں کیا صورتیں ہوں گی کہ پنہاں ہو گئیں" اس شعر میں کون سی صنعت استعمال ہوئی ہے۔

حصہ سوم

سوال نمبر 3: (الف) ارکان استعارہ شعری مثال کے ساتھ بیان کریں۔

05

(ب) زحاف کی وضاحت کریں۔

05

سوال نمبر 4: مجاز مُرسل سے کیا مراد ہے؟ اس کی مختلف صورتیں وضاحت سے لکھیں۔

10

سوال نمبر 5: ا مندرجہ ذیل شعر کی تقطیع کریں اور بحر کا نام تحریر کریں۔

10

غلامی میں نہ کام آتی ہیں شمشیریں نہ تدبیریں

جو ہو ذوق یقین پیدا تو کٹ جاتی ہیں زنجیریں



درست جوابات : معروضی حصہ

الف	1
ج	2
ب	3
ب	4
ب	5
ج	6
الف	7
ب	8
ب	9
ج	10

**NATIONAL UNIVERSITY OF PAKISTAN**

BS Program (Model Paper)
End-Semester Examination

Reg.No: _____

Course Code: MTD-112

Course Title: Pre-Mathematics-II

Sign of Invigilator: _____

Time Allowed: 20 minutes

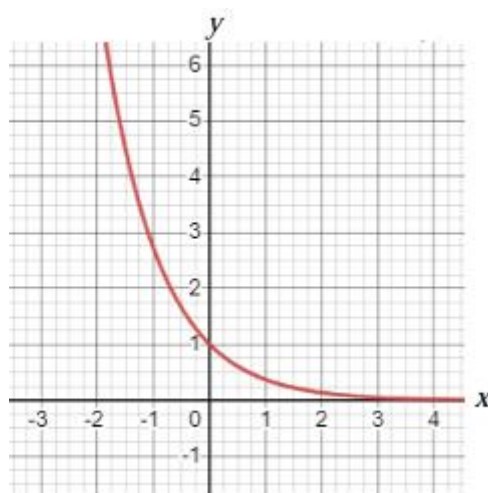
Total Marks: 10

General Instructions

- Deleting/Overwriting is not allowed. Do not use lead pencil.
- All parts of objective section are to be answered on this paper and handed in to the invigilator.

SECTION – A**Q1. Choose the most appropriate option.****(10*1=10)**

1. A function is represented by the given graph

The range of $f(x)$

will be

- | | |
|------------------------|------------------|
| A. $(-\infty, \infty)$ | C. $(0, \infty)$ |
| B. $(-\infty, 0)$ | D. $(1, \infty)$ |
2. If $f(x) = \sin(\pi - x)$, then $f'(x)$ will be
- | | |
|---------------------|-------------------------------|
| A. $-\cos(\pi - x)$ | C. $-(\pi - x) \cos(\pi - x)$ |
| B. $\cos(\pi - x)$ | D. $(\pi - x) \cos(\pi - x)$ |
3. The function $f(x) = x^2 - 2x - 3$ has an extreme value at
- | | |
|---------|--------|
| A. -1 | C. 2 |
| B. 1 | D. 3 |
4. $\int \ln(\sqrt{x}) \times \frac{1}{x} dx$ is equal to
- | | |
|----------------------------------|---------------------------------|
| A. $2[\ln(\sqrt{x})]^2 + C$ | C. $\frac{\ln x}{\sqrt{x}} + C$ |
| B. $\frac{\ln x}{2\sqrt{x}} + C$ | D. $[\ln(\sqrt{x})]^2 + C$ |



5. The point $(1, x)$ divides the line segment joining $(-2, -1)$ and $(2, 3)$ in the ratio 3:1. The value of x will be
- A. -2 C. -1
B. 1 D. 2
6. The distance between two parallel lines $x - y - 1 = 0$ and $x - y - 2 = 0$ is
- A. 1 unit C. $\frac{1}{\sqrt{2}}$ units
B. $\sqrt{2}$ units D. 2 units
7. The joint equation for the two given equations of straight lines $x + 2y = 0$ and $y = x$ will be
- A. $2x + 3y = 0$ C. $xy + 2y^2 = 0$
B. $x^2 + 2y^2 = 0$ D. $x^2 + xy - 2y^2 = 0$
8. The centre of the circle $x^2 + y^2 + 8x + 18y + 5 = 0$ is
- A. $(4, 9)$ C. $(-4, -9)$
B. $(8, 18)$ D. $(-8, -18)$
9. A garden has an elliptical walking track around it. On the walking track, the maximum distance is $20m$. Assume that, the minimum distance between a person sitting at the centre and a person standing on the walking track is $6m$.



The equation that represents the elliptic path will be

- A. $\frac{x^2}{100} + \frac{y^2}{6} = 1$ C. $\frac{x^2}{100} + \frac{y^2}{36} = 1$
B. $\frac{x^2}{200} + \frac{y^2}{36} = 1$ D. $\frac{x^2}{200} + \frac{y^2}{6} = 1$
10. The order of the differential equation $x^3 \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} = 0$ is
- A. 0
B. 1
C. 2

**NATIONAL UNIVERSITY OF PAKISTAN****BS Program (Model Paper)****End-Semester Examination**

Reg.No: _____

Course Code: MTD-112**Time Allowed:** 02 Hrs and 40 minutes**Course Title:** Pre-Mathematics II**Total Marks:** 40**Sign of Invigilator:** _____**General Instructions**

- Do not write anything on the question paper other than Reg.No.
- Attempt all questions. All the questions & answers must be numbered as mentioned in the question paper.

SECTION – B**Q.2 Provide a comprehensive solution to the following questions: (5*2=10)**

1. Solve each of the given parts

a. If $f(x) = 3x^2 - 5x + 1$ and $g(x) = x + 1$, then find $g \circ f$.b. Find limit of $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{3n}\right)^n$ in terms of e .2. Find $\frac{dy}{dx}$ for the given parametric equations $x = \sin t$, $y = \cos t$.

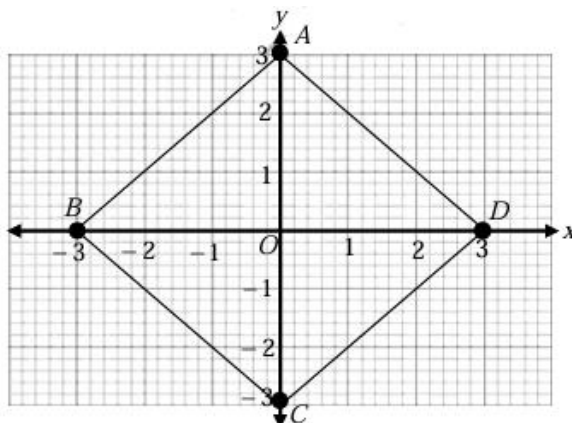
3. Divide 20 into two parts so that the sum of the squares will be minimum.

4. Find the area of

quadrilateral $ABCD$ with the given vertices.

(week: 10-

11)

5. Find the equation of normal to the parabola $y^2 = -8x$ at $(2, 2)$ in the standard form.

**SECTION – C**

Write detailed answers to the following questions.

(10*3=30)

Q.3. Solve both parts, each carry equal marks

- a. For the piecewise function $f(x) = \begin{cases} x + 2 & \text{if } x \leq -1 \\ c + 2 & \text{if } x > -1 \end{cases}$ find "c" so that limit exists.

Also draw the graph of the function.

- b. Find derivative of the function $f(x) = \log_a x$ by definition.

Q.4. Solve both parts, each carry equal marks

a. $\int x \tan^2 x \, dx$

- b. An object is dropped from the top of the tower, vertically downward, under the influence of gravitational force. Taking gravitational acceleration as 10 m/s^2 , find

- Expression of the velocity v after t seconds of falling.
- Velocity after two seconds of the fall.

Q.5. Solve both parts, each carry equal marks

- The vertices of a hyperbola are $(15, 1)$ and $(-1, 1)$. If the length of its conjugate axis is 12 units. Then find the equation of the hyperbola in standard form.
- Solve the given non-linear equation $f(x) = x^3 - 4x - 9$ numerically using **Newton-Raphson Method**

**Answer Key**

QUESTION NUMBER	ANSWER
1	C
2	A
3	B
4	D
5	D
6	C
7	D
8	C
9	C
10	C



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